The Mechanics of Delta Learning: Target Design for Generalizable Scientific Machine Learning

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Abstract

In scientific machine learning, Δ-learning trains models on residual errors relative to approximate baselines, assuming that smaller residual scales inherently yield better downstream accuracy. Here, we demonstrate that residual scale alone is an insufficient heuristic for learnability. For molecular total energies, replacing a linear baseline with a neural network fitted to the same elemental counts reduced the residual scale, yet more than doubled GemNet's prediction error. Across local structural descriptors, more flexible baselines produced smaller residuals that varied sharply between structurally similar molecules. By contrast, semi-empirical baselines left larger residuals, yet yielded lower test error and improved out-of-domain transferability. Across the targets studied, scale-normalized graph Dirichlet roughness tracked scale-normalized prediction error, suggesting its use as a pre-training diagnostic for target selection. These results motivate *baseline complementarity* as a target-design principle in scientific machine learning: a useful baseline captures contributions the downstream architecture struggles to predict.

1 Introduction

Supervised machine learning learns a mapping from an input representation to a target property. The generalizability of this mapping fundamentally depends on its smoothness—the degree to which continuous, localized shifts in feature space produce smooth variations in the target. Conventional model development approaches this problem primarily from the representation side, refining neural architectures and inductive biases to construct feature spaces where the target varies smoothly [1, 2, 3, 4, 5]. The prediction target itself, by contrast, is usually treated as fixed.

Yet heterogeneous targets, such as molecular total energies, aggregate compositional, local atomic, many-body, and long-range electronic interactions that span diverse physical scales. A given graph neural network (GNN) architecture may represent these distinct physical contributions with varying fidelity. When a component of the target aligns poorly with a model's intrinsic representation, it increases target roughness within that feature space, placing a fundamental bottleneck on downstream generalizability.

Δ-learning is conventionally motivated by a simple heuristic: subtracting a more accurate physical baseline leaves a smaller residual, thereby easing the downstream prediction task [6, 7, 8, 9, 10]. In practice, however, this assumption frequently breaks down [11]. For example, in a study of isoelectronic diatomics, Krug *et al.* [12] demonstrated that while their proposed baseline reduced the training data required for chemical accuracy by up to an order of magnitude, an alternative baseline combining an alchemical harmonic approximation with a Morse potential yielded negligible gains—and occasionally worsened predictions compared to direct target learning. These contrasting outcomes expose a central, unanswered question in scientific AI: *what does subtracting a baseline alter about the learning task beyond simply shrinking the target's scale?*

Subtracting a baseline simultaneously alters two distinct properties of the target: its **scale** and its **functional form**. Holding scale-normalized learnability constant, a smaller residual directly yields a lower absolute prediction error. However, altering the functional form can cause the residual to vary more sharply between inputs that the network represents similarly—effectively increasing prediction error relative to target scale. Δ-learning must therefore be reframed as an active problem of *target design*, where candidate baselines are evaluated by their dual ability to compress target scale while preserving or enhancing smoothness within the downstream architecture's feature space.

To decouple intrinsic learnability from target scale post-training, we define the scale-normalized mean absolute error (n-MAE) as the prediction MAE divided by the target interquartile range (IQR). An increase in n-MAE reveals that a residual target has become intrinsically harder for the architecture to resolve relative to its spread. The ideal baseline transformation reduces overall target scale without incurring an n-MAE penalty.

Because n-MAE can only be measured after model training, rational target design requires a pre-training diagnostic of feature-space smoothness. Here, we introduce IQR-normalized graph Dirichlet roughness (D_{IQR}), a metric that directly quantifies target variation across fixed, architecture-informed feature spaces prior to training [13]. A low D_{IQR} value indicates that molecules positioned close together in feature space exhibit consistent target values relative to scale, providing an *a priori* screening tool for candidate baselines.

We systematically evaluate this target-design framework across molecular total energy predictions using GemNet-T and SchNet [14, 4], comparing linear descriptors, non-linear machine-learned fits, and two semi-empirical quantum-mechanical baselines, GFN2-xTB [15] and density-functional tight binding (DFTB). In-domain accuracy is benchmarked on organometallic complexes from tmQM, and out-of-distribution transferability is tested on organic molecules from QM9 [16, 17]. Across both architectures, replacing simple linear baselines with flexible non-linear fits yields smaller but rougher residuals and higher normalized error. The xTB baseline reduces target scale and normalized roughness and improves both in-domain accuracy and out-of-domain transferability.

Ultimately, these findings frame Δ-learning as a two-dimensional optimization problem over **residual scale** and **representation-space smoothness**, establishing target-space formulation alongside model architecture as a fundamental axis of scientific machine learning.

2 Results

2.1 Baseline flexibility at fixed information

To test whether minimizing residual scale inherently improves downstream accuracy, we compare baselines that differ in flexibility but use the same elemental counts. Following standard practice [18, 4, 19], we subtract fixed element-wise atomic references from the total DFT energy to obtain the reference-corrected energy (E_{ref}). We then fit a linear Bag-of-Elements (BoE) model to atomic counts and subtract its prediction (E_{BoE}) to construct the residual target:

$$E_{\mathrm{ref,BoE}} = E_{\mathrm{ref}} - E_{\mathrm{BoE}} \tag{1}$$

Because the subtracted baseline is restored at inference during stacked Δ-learning, the mean absolute error (MAE) of the residual prediction is identically equal to that of the complete pipeline (SI Section 1).

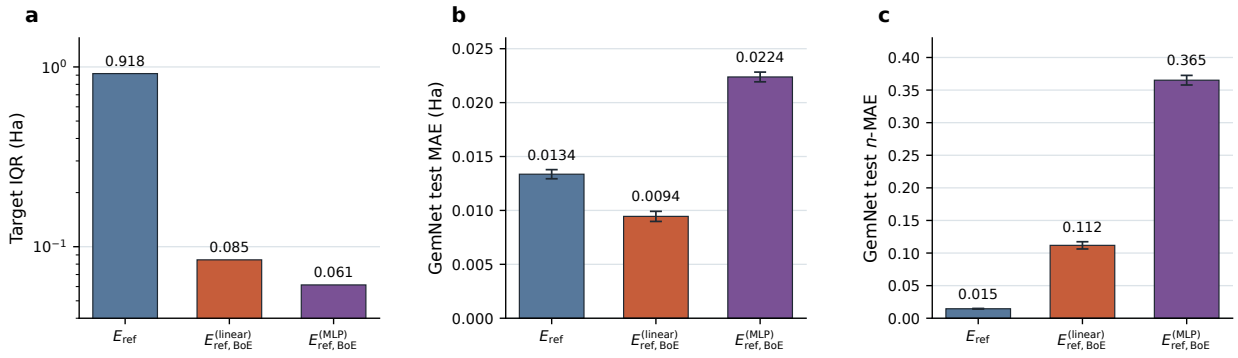


Figure 1: **Target scale and GemNet accuracy under linear and non-linear BoE corrections. a,** Test-set interquartile range (IQR) for the reference energy E_{ref} and its linear- and MLP-BoE residuals on tmQM. **b,** Mean GemNet test MAE across targets. **c,** Scale-normalized test error (n-$\mathrm{MAE}_{\mathrm{test}}$). Error bars represent sample standard deviations across three independent training seeds.

We report the test mean absolute error ($\mathrm{MAE}_{\mathrm{test}}$) and its scale-normalized counterpart,

$$n\text{-}\mathrm{MAE}_{\mathrm{test}} = \frac{\mathrm{MAE}_{\mathrm{test}}}{\mathrm{IQR}_{\mathrm{test}}(y)} \tag{2}$$

The normalized error expresses the prediction error relative to the spread of the test target and is invariant to changes in energy units (SI Note 2).

Evaluating GemNet-T on the tmQM dataset reveals a trade-off between target scale and learnability (Fig. 1). Applying the linear BoE correction compresses the target IQR by a factor of 10.9, but increases GemNet's normalized error by a factor of 7.7 (Fig. 1a,c). Although the residual becomes harder for GemNet to learn relative to its spread, the reduction in scale outweighs the increase in normalized error, yielding a 1.41-fold net reduction in absolute test MAE (Fig. 1b).

This gain in absolute accuracy motivates a question: *does pushing residual scale even smaller further improve predictions?* To test this without injecting extra structural information, we replace the linear baseline with a more flexible multilayer perceptron (MLP) trained on the same elemental counts. The MLP baseline reduces the residual spread by an additional factor of 1.38 relative to the linear model. However, GemNet's normalized error increases by a factor of 3.27 (Fig. 1a,c).

The increase in normalized error outweighs the reduction in target scale, causing GemNet's final test MAE to more than double (2.37×) compared to the linear baseline (Fig. 1b). SchNet shows the same trend (SI Section 1). Across seeds, both architectures achieved lower mean final-epoch training MAE on the MLP-BoE residual than on the linear-BoE residual, yet their best validation and corresponding test MAEs were higher (Supplementary Table S3).

These results demonstrate that residual scale alone does not predict downstream accuracy. Because normalized error can only be measured post-training, selecting effective baselines requires an *a priori* diagnostic of target learnability. To evaluate candidate targets prior to model training, we next investigate how smoothly residual values vary across molecular structural representations.

2.2 Pre-training roughness tracks target learnability *a priori*

To evaluate candidate residuals before downstream training, we measure how target values vary across fixed geometric feature spaces built to serve as **representation proxies** for the downstream architectures. The **Pair-Radial Expansion (PRE)** describes molecular geometry through pairwise distance distributions, acting as a proxy for SchNet's pairwise representation. **PRE+A** extends PRE by incorporating central-element triplet angle distributions, capturing the directional and angular information accessible to GemNet-T. (Both proxy spaces are later reused as baseline models.)

Within each proxy space, we construct a k-nearest-neighbor graph connecting test-set molecules with similar structural representations and compute the scale-normalized graph Dirichlet roughness, $D_{\mathrm{IQR}}(y)$ (Methods, Section 5.4, Equation 7). This metric measures target variation between neighboring molecules relative to the central target spread (IQR), where lower scores indicate that geometrically similar molecules possess more consistent target values. Applied to the composition-corrected targets, D_{IQR} resolves the learnability paradox from Figure 1: in the GemNet-T proxy space (PRE+A), normalized roughness increases from 0.548 for E_{ref} to 1.127 for the linear-BoE residual, surging to 1.658 for the MLP-BoE residual (and 1.709 vs 1.152 in the SchNet proxy space, PRE; SI Note 6). Pre-training target roughness thus directly tracks GemNet-T's post-training normalized error (n-MAE$_{\mathrm{test}}$), demonstrating that non-linear composition baselines shrink target scale only by injecting fine-grained roughness into the target landscape.

2.3 Residual roughness guides target design beyond baseline accuracy

For baselines fitted to elemental counts, normalized roughness tracked GemNet's error relative to target scale. We next test whether this relationship extends to baselines that incorporate molecular structure. We fit linear and MLP baselines to three feature families of increasing structural detail (Table 1): the Bag-of-Bonds (BoB) descriptor (counting bonds by type), along with PRE and PRE+A—the geometric representation proxies introduced in §2.2, which now serve directly as inputs to the baseline models.

Table 1: **Baseline models used for target transformation.**

Baseline model	Input features	Fitting approach	Physical/representation target
BoB	RDKit bond-type counts rather than conventional BoB Coulomb-matrix entries	Linear / MLP	Compositional and functional-group variation
PRE	Pairwise distances (r_{ij})	Linear / MLP	Local two-body geometry (SchNet proxy)
PRE+A	Distances and angles (r_{ij}, θ_{ijk})	Linear / MLP	Local three-body geometry (GemNet proxy)
xTB	Atomic identities and molecular coordinates	Semiempirical QM	Non-local electronic and higher-order interactions
DFTB†	Atomic identities and molecular coordinates	Semiempirical QM	Molecule-wide electronic structure; supporting comparison with xTB

All three descriptor families use the same target: the atom- and composition-referenced DFT energy, $E^{\mathrm{DFT}}_{\mathrm{ref,BoE}}$. For each fitted baseline b, the downstream GNN learns the residual difference:

$$\Delta^b = E^{\mathrm{DFT}}_{\mathrm{ref,BoE}} - \widehat{E}_b, \tag{3}$$

This experimental setup separates two distinct effects. Comparing BoB, PRE, and PRE+A at a fixed MLP architecture tests what the baseline gains from additional structural information; comparing linear and MLP fits within the same descriptor family tests the effect of baseline flexibility while holding input information constant.

For the physical baseline comparison, we use the energy from a semi-empirical xTB calculation [15], applying the same referencing protocol before taking the residual difference:

$$\Delta^{\mathrm{xTB}} = E^{\mathrm{DFT}}_{\mathrm{ref,BoE}} - E^{\mathrm{xTB}}_{\mathrm{ref,BoE}}. \tag{4}$$

Adding structural detail reveals the same flexibility penalty. As expected, incorporating structural information improves the MLPs' native predictions: baseline error falls from BoB to PRE to PRE+A, producing residuals narrower than the DFT parent by factors of 3.3, 4.2, and 4.7 in test IQR (Fig. 2a,c). Yet all three residuals vary more sharply between molecules that are close in the GemNet proxy space (Fig. 2b), and none improves GemNet's test MAE over direct training on the DFT parent (Fig. 2c). Thus, the descriptor baseline that best approximates DFT energies leaves the smallest target, but that target does not yield the most accurate final prediction.

The same pattern holds when descriptor complexity is held fixed. Within each family, replacing the linear fit with an MLP produces a smaller, rougher residual and increases GemNet's test error. For example, comparing the two BoB baselines, the MLP reduces the residual IQR by a further factor of 1.6, yet GemNet's test MAE rises by a factor of 1.7 (Fig. 2). The trade-off observed with elemental counts in §2.1 therefore persists across structured representations.

By contrast, the semi-empirical xTB baseline yields higher accuracy. Although xTB leaves a larger residual than any of the three descriptor-based MLPs, GemNet predicts it more accurately. Relative to the DFT parent, normalized roughness falls from 1.127 to 0.964, and mean GemNet test MAE drops from 0.00945 to 0.00660 Ha (Fig. 2b,c). This advantage persists after accounting for target scale: GemNet's normalized test error is 7.2 times higher on the MLP-PRE residual than on the xTB residual (and 4.9 times higher for SchNet; SI Note 6). These results align with baseline complementarity—akin to observations by Tu and Rowley [20], where a DFTB3 baseline

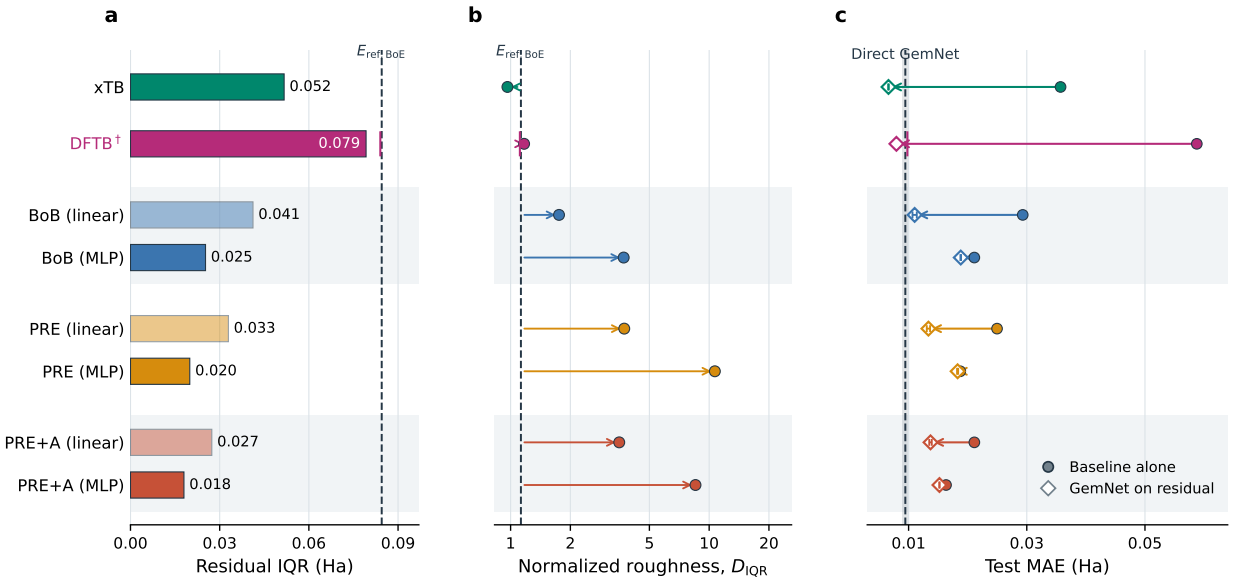


Figure 2: **Baseline accuracy, residual roughness, and GemNet test performance. a,** Test residual IQR. **b,** Normalized roughness on the test PRE+A graph (log scale). **c,** Baseline-only test MAE (filled circles) and GemNet residual test MAE (open diamonds). Dashed lines mark the DFT-BoE parent or direct-GemNet reference; arrows show the corresponding changes. Main GNN results are three-seed means; error bars and shading show sample SD. Magenta DFTB† is a single-seed, valid-subset control, excluded from pooled analyses. Short magenta ticks mark its matched-subset parent references (SI Note 8).

supplied long-range electrostatic contributions beyond a local network's cutoff to improve interaction energies.

Finally, the DFTB control [21, 22] illustrates why absolute accuracy and normalized roughness must still be distinguished. Because DFTB calculations failed for 362 neutral tmQM molecules, we evaluated one GNN seed per architecture on the successful subset (excluding this control from pooled analyses; SI Note 8). On that matched subset, GemNet test MAE fell from 0.00979 to 0.00794 Ha even though normalized roughness in PRE+A space rose from 1.113 to 1.169. While lower roughness accompanied the accuracy gains for xTB, the DFTB comparison shows that absolute error can improve even when normalized roughness increases.

2.4 Model-relative target roughness tracks scale-normalized learnability

Because D_{IQR} and n-MAE share target IQR as a scaling factor, part of their strong correlation reflects shared-denominator normalization [23]; nevertheless, joint-label permutation tests confirm that architecture-aligned target roughness tracks relative learnability beyond this scaling effect. Across all 13 primary targets, GemNet's scale-normalized test error (n-$\mathrm{MAE}_{\mathrm{test}}$) increases approximately linearly with log PRE+A Dirichlet roughness, yielding $R^2 = 0.905$ (Fig. 3; SchNet exhibits $R^2 = 0.946$ in PRE space; SI Note 6). While unnormalized metrics show no such trend (SI Note 7), shuffling target values among molecules preserves target IQRs while destroying structural alignment; across 5,000 joint-label permutations, none produced a positive correlation as high as the original molecular assignment (median shuffled $R^2 = 0.862$; one-sided $p \approx 0.0002$ [24]; SI Note 7). These retrospective results establish D_{IQR} as a scale-invariant pre-training diagnostic of relative target

learnability. Its ability to guide baseline selection for new target families remains to be tested.

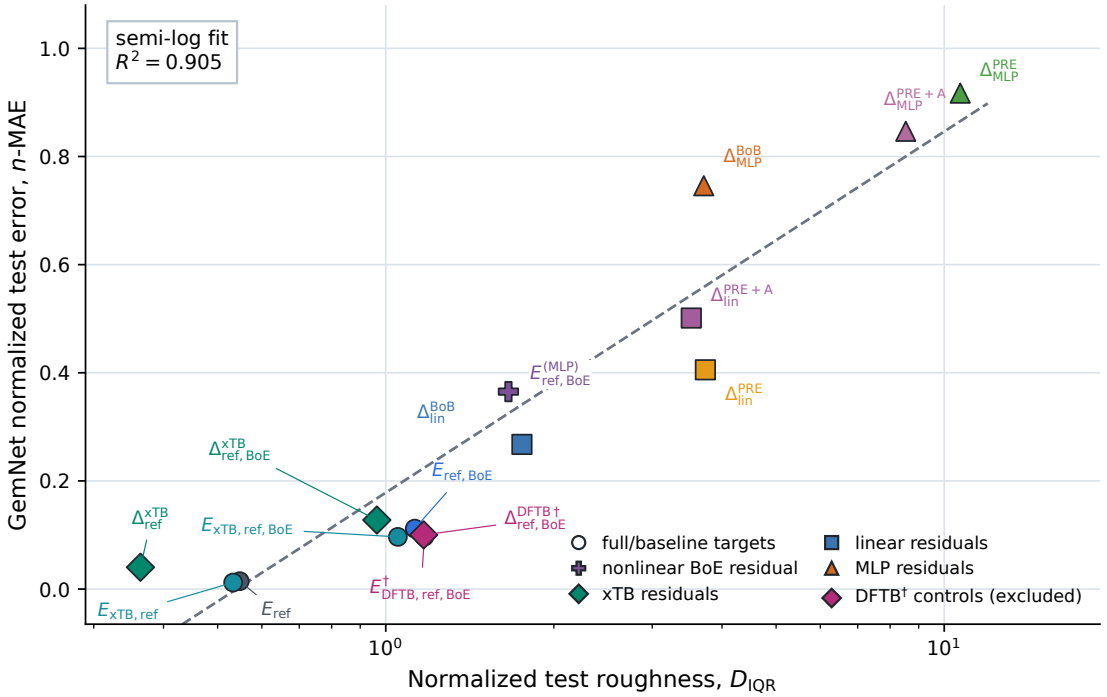


Figure 3: **Model-relative roughness and normalized GemNet test error.** Mean GemNet test n-MAE, defined as $\mathrm{MAE}_{\mathrm{test}} / \mathrm{IQR}_{\mathrm{test}}(y)$, is plotted against test D_{IQR} for 13 primary targets. Roughness is measured on the standardized test PRE+A graph, and the horizontal axis is logarithmic. Marker shapes identify the target classes shown in the legend. Points show means across three GNN seeds. The dashed line is the semi-log regression over the 13 primary target means ($R^2 = 0.905$). Magenta DFTB† points are single-seed, valid-subset controls, excluded from the regression and correlations.

2.5 Target design governs out-of-domain transferability

To test whether baseline choice affects transfer across chemical domains, we evaluated models trained on tmQM transition-metal complexes using small organic molecules from QM9 [17]. We recomputed QM9 reference energies at the identical level of DFT theory and kept all baseline parameters fitted on tmQM fixed. Because out-of-domain (OOD) data reside in a shifted energy manifold, models inevitably suffer from global miscalibration, rendering absolute error metrics like raw MAE sensitive to dataset-wide shifts rather than local predictive fidelity. Because relative energy ordering and potential energy surface topography govern chemical phenomena such as structural stability and reaction energetics, trend preservation is essential for assessing whether a model has learned transferable chemical physics rather than domain-specific offsets. To rigorously assess this trend preservation, we evaluate model predictions using shift-independent metrics: Kendall rank correlation (τ) to measure energetic ordering regardless of global shifts, and mean-shifted MAE to quantify residual scatter after removing a single dataset-wide offset estimated from QM9 reference labels (SI Note 5).

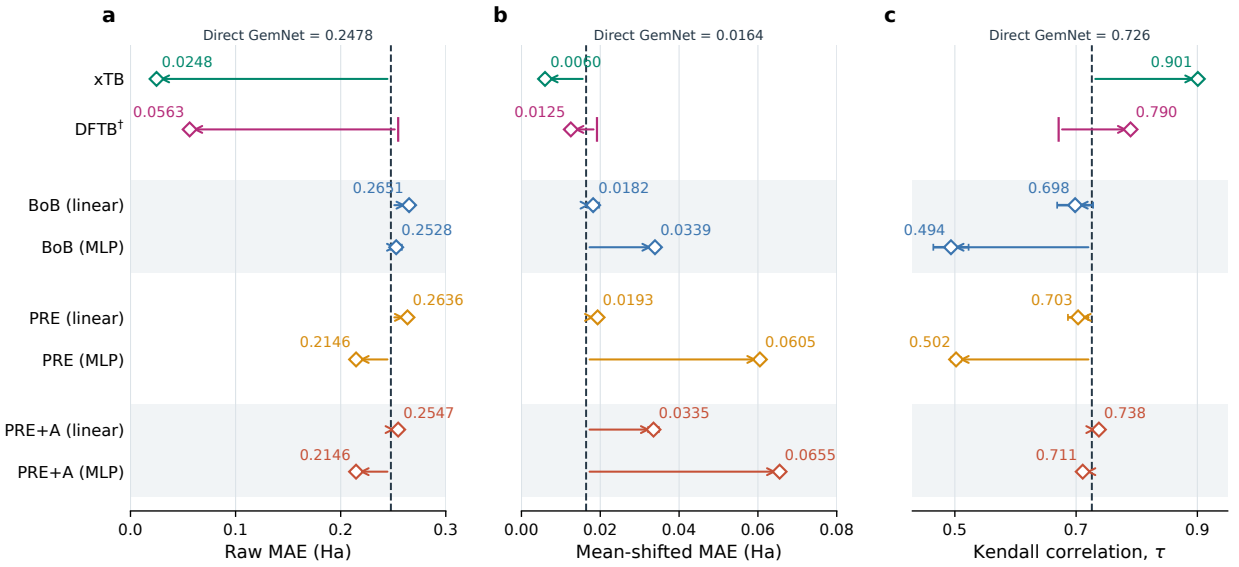


Figure 4: **Out-of-domain GemNet transfer from tmQM to QM9. a,** Raw MAE. **b,** Mean-shifted MAE using one offset estimated from QM9 labels. **c,** Kendall correlation. Diamonds show baseline-plus-GemNet predictions against $E_{\mathrm{ref,BoE}}$; dashed lines mark direct GemNet. Primary results are three-seed means with sample-SD bars on 128,988 common molecules. Magenta DFTB† is a seed-101 control on 128,978 valid molecules; its short reference ticks and arrows use direct seed 101 on the same subset (SI Notes 5 and 8).

On the common QM9 cohort, standalone xTB achieved a raw MAE of 0.02666 Ha, a mean-shifted MAE of 0.01809 Ha, and Kendall $\tau = 0.677$. Adding the GemNet correction reduced raw MAE modestly to 0.0248 Ha, but lowered mean-shifted MAE to 0.00598 Ha and raised τ to 0.901 (Fig. 4). Direct GemNet further illustrates why these metrics are necessary: its raw MAE was 0.2478 Ha—more than nine times that of xTB—yet its higher Kendall score ($\tau = 0.726$) shows that GemNet captured underlying chemical trends better but suffered from a severe global offset. Correcting for this offset reveals a mean-shifted MAE of 0.01641 Ha, outperforming standalone xTB. Thus, xTB provided baseline calibration, while the learned correction substantially refined molecular energy ordering and reduced residual scatter. SchNet exhibited parallel improvements, raising Kendall τ from 0.746 (direct learning) to 0.895 (xTB residual; SI Note 5).

By contrast, descriptor-based MLPs failed to improve transferability. No local descriptor baseline surpassed direct GemNet across both trend metrics; for example, linear PRE+A increased mean-shifted MAE to 0.03351 Ha despite a minor gain in Kendall τ (0.738). DFTB provided a second test of electronic-structure baseline complementarity: on its valid subset, DFTB + GemNet reduced raw MAE from 0.2547 to 0.0563 Ha, lowered mean-shifted MAE from 0.01917 to 0.01253 Ha, and increased Kendall τ from 0.671 to 0.790 relative to direct GemNet (Fig. 4; SI Notes 5 and 8).

These OOD results reinforce the principle of baseline complementarity. Local geometric descriptors mirror message-passing GNN inductive biases, stripping away smooth local features and yielding rough, difficult targets. Conversely, semi-empirical quantum methods capture long-range electrostatics and charge redistribution that local networks represent poorly. By preserving or enhancing scale-normalized representation smoothness (D_{IQR}) while shrinking target scale, complementary baselines systematically support both in-domain generalization and out-of-domain transferability.

3 Discussion

In scientific machine learning, Δ-learning rests on a compelling intuition: rather than predicting a high-fidelity quantum-mechanical energy from scratch, a GNN should learn the correction to a cheap approximate baseline. Because the baseline captures much of the variation in the target, the resulting residual is smaller and presumed easier to learn. Yet, as our findings demonstrate, this intuition does not always hold. Aggressively shrinking total target scale does not guarantee downstream learnability, explaining why Δ-learning has often relied on trial-and-error heuristics rather than design principles.

Our results resolve this inconsistency by reframing Δ-learning as a functional target transformation governed by two distinct properties: **target scale** and **scale-normalized representation smoothness**, which tracks downstream learnability as assessed by n-MAE. Because absolute test error is the product of target scale and normalized error, aggressive scale reduction can still yield a net accuracy gain if the shrinkage outweighs the accompanying loss in learnability—as seen with the linear Bag-of-Elements (BoE) baseline. However, pushing shrinkage even further with a nonlinear MLP-BoE baseline raises normalized error enough to outweigh the further reduction in target scale and degrades performance.

Optimal Δ-learning therefore requires **baseline complementarity**: we propose prioritizing baselines that capture the specific components responsible for the model's error, rather than merely absorbing the largest share of total target spread. Local descriptor baselines (Bag-of-Bonds, PRE, and PRE+A) share geometric information with local message-passing GNNs; they strip away smooth local features the network already represents efficiently, leaving residuals that are narrower in scale yet disproportionately rough and harder to learn, resulting in poorer test performance than direct learning. Conversely, semi-empirical quantum methods (xTB and DFTB) capture electronic effects—such as charge redistribution and long-range electrostatics—that may contribute to the errors of local architectures. By absorbing the specific energy contributions that drive GNN error, complementary baselines shrink target scale while preserving or reducing scale-normalized roughness, improving test accuracy by 30–40% and substantially enhancing energetic rank correlation across distinct chemical domains (QM9).

Because post-training evaluation requires expensive model optimization, practical target engineering requires pre-training diagnostics. To address this, we introduced the scale-normalized graph Dirichlet roughness (D_{IQR}), which tracks target variation over architecture-aligned representation proxies without training a downstream GNN. Its strong semi-log association with normalized test error (n-MAE) supports D_{IQR} as a candidate pre-training diagnostic for screening baselines. While shared normalization contributes substantially to this association, joint-label permutation controls support an additional contribution from molecular-neighbourhood alignment.

4 Conclusion

This study advances Δ-learning from heuristic baseline selection toward principled target design. By establishing that baseline complementarity and the balance between target shrinkage and residual smoothness govern learnability, our results position target formulation as a foundational design choice alongside network architecture, offering a clear path forward for residual learning, multi-fidelity modeling, and scientific machine learning surrogates.

5 Methods

5.1 Datasets and targets

The in-domain data are the aligned neutral tmQM transition-metal complexes with ωB97M-V/def2-SVPD energies and fixed train, validation, and test molecule identifiers [16, 25, 26]. The full panel contains 56,830 training, 7,104 validation, and 7,104 test molecules. The six descriptor residuals use 56,821 training and 7,103 validation molecules, while all 13 primary targets share the same 7,104 test molecules. Both architectures use identical target values and split memberships within each target. DFTB success filtering leaves 56,540 training, 7,068 validation, and 7,068 test molecules; failed calculations were not imputed. QM9 is used only for OOD evaluation; its energies were recomputed at the same level with Psi4 to avoid conflating chemical-domain shift with electronic-structure mismatch [17, 27].

For method m, molecular energies were first corrected using fixed element-wise atomic references,

$$E^m_{\mathrm{ref}}(i) = E^m_{\mathrm{raw}}(i) - \sum_Z n_Z(i)\epsilon^m_Z, \tag{5}$$

then corrected by a linear model of elemental counts fitted only on the tmQM training split to obtain $E^m_{\mathrm{ref,BoE}}$. Atomic references are fixed external inputs, not parameters fitted to held-out molecular energies. Separate composition models are fitted for DFT, xTB, and DFTB and frozen for validation, test, and QM9 evaluation. The DFT quantity is the parent target. Delta targets subtract either a fitted descriptor baseline or the consistently referenced xTB/DFTB energy. Full target-generation, masking, and molecule-ID alignment details are in SI Methods.

5.2 Baseline models

BoE denotes elemental-count features and the standard BoE reference is linear. A composition-only control fits a nonlinear MLP to the same counts and atom-referenced DFT energy, thereby changing baseline flexibility without adding structural information. BoB denotes the study-specific RDKit-derived typed-bond-count vector, not the conventional Coulomb-matrix Bag-of-Bonds descriptor. PRE expands local element-resolved pair distances in Gaussian radial basis functions up to a spatial cutoff, and PRE+A augments PRE with central-element-resolved triplet angle distributions. BoB, PRE, and PRE+A each have linear ridge and MLP fits to the DFT-BoE parent target; all four MLP baselines share hidden widths of 512 and 128 and the same training protocol. Ridge regularization was selected by five-fold training-only cross-validation, with preprocessing refitted within each fold. Final baseline parameters and preprocessing were fitted using only the corresponding tmQM training split. The validation split was used to select each MLP checkpoint, while test labels were used only for final evaluation. Baseline fitting seed 101 defines the canonical residual targets; fitting-seed sensitivity is reported separately in SI Note 9. GFN2-xTB and DFTB are fixed semiempirical baselines [15, 21, 22]. DFTB uses SCC-DFTB/DFTB2 with the PTBP Slater–Koster set, without third-order or dispersion corrections [28], and only paired successful calculations; failures were not imputed. Feature dimensions, hyperparameters, and failure audits are in SI Methods and SI Note 8.

5.3 GNN training

GemNet-T is the primary downstream model and SchNet is the architecture control [14, 4]. Models were trained on energies only, without forces or periodic boundary conditions, using fixed splits, a 6.0 Å cutoff, and a common FairChem/OCP-style protocol. Every target was standardized by its own training-set mean and population standard deviation. Within each architecture, architecture

settings, loss, AdamW configuration, initial learning rate, scheduler rule, batch size, and checkpoint criterion were held fixed across targets. AdamW with AMSGrad used an initial learning rate of 5×10^{-4}, MSE energy loss, and an 80-epoch budget; batch sizes were 16 for SchNet and four for GemNet. ReduceLROnPlateau responded independently to each validation trajectory, with factor 0.8 and patience three, producing target-specific learning-rate schedules. Each run's best-validation checkpoint was evaluated on both the tmQM test set and QM9. All 13 primary targets use independent GNN seeds 101, 102, and 103 for both architectures; the two DFTB targets use seed 101 only for both architectures, giving 82 runs in total. Reported primary results are means of seed-level metrics, not ensemble predictions, with sample standard deviations shown where applicable. Single-seed DFTB results have no across-seed uncertainty estimate. Exact seed-level results are reported in SI Note 9. Complete configurations and training diagnostics are in SI Methods and Supplementary Table S3.

5.4 Roughness and transfer diagnostics

GemNet roughness uses standardized PRE+A features and SchNet roughness uses standardized PRE features. A cosine $k = 10$ nearest-neighbour graph was built on the 7,104-molecule tmQM test cohort. Zero-variance feature channels were removed, and retained channels were divided by the graph cohort's population standard deviation without centring. This unsupervised graph preprocessing is distinct from train-only baseline preprocessing. Unique undirected union edges were assigned similarity-based weights using the locally scaled Gaussian kernel in Equation 6.

$$w_{ij} = \exp\left(-\frac{d_{ij}^2}{\sigma_i \sigma_j}\right), \tag{6}$$

Here, d_{ij} is the cosine distance between standardized descriptor vectors for molecules i and j, and σ_i is a local distance scale, defined as the distance from molecule i to its k-th nearest neighbour.

Using these edge weights, we quantify target variation through the IQR-normalized graph Dirichlet roughness, D_{IQR} [13]:

$$D_{\mathrm{IQR}}(y) = \frac{1}{\sum_{\{i,j\}\in\mathcal{E}} w_{ij}} \sum_{\{i,j\}\in\mathcal{E}} w_{ij} \frac{(y_i - y_j)^2}{\mathrm{IQR}_{\mathrm{test}}(y)^2}. \tag{7}$$

Here, $\mathcal{E}$ is the set of unique undirected graph edges. For targets with nonzero IQR, this formulation normalizes local squared target differences by the squared central target spread, making D_{IQR} invariant to nonzero uniform target rescaling.

Both D_{IQR} and n-$\mathrm{MAE}_{\mathrm{test}}$ use the corresponding test-target IQR; graphs and weights remain fixed across primary targets within each architecture. DFTB controls use separately constructed valid-subset test graphs and are excluded from primary regressions and pooled statistics.

Semi-log regressions use unweighted ordinary least squares across the 13 primary target means, with an intercept and $\ln D_{\mathrm{IQR}}$ as predictor. Raw-error, raw-roughness, variance-normalization, partial-correlation, and leave-one-target-out sensitivity analyses are reported in SI Note 7. The same note reports 5,000 joint test-label permutations, which preserve each target's distribution and IQR while breaking molecular-neighbourhood alignment, and comparisons of target pairs whose IQRs differ by at most 10%. Leave-one-target-out fits are influence checks, not cross-validated prediction scores. These test-set analyses are retrospective diagnostics performed after model selection; test labels were not used to fit baselines, schedule training, or select checkpoints.

5.5 OOD transfer evaluation

tmQM-trained models were evaluated without retraining on QM9 geometries, using frozen tmQM training-only baseline parameters. Residual predictions were combined with their corresponding baselines and mapped to the common DFT-BoE parent before full-target correlation and ranking metrics were calculated. Primary parent-energy comparisons use 128,988 common QM9 molecules and three GNN seeds. The DFTB residual is shown on its 128,978-molecule intersection with this cohort, alongside a matched-subset direct seed-101 reference; this matches evaluation molecules, not training cohorts. Mean-shifted MAE removes one scalar mean-error offset estimated separately for each predictor from the evaluated QM9 reference labels and is used only as a post hoc oracle diagnostic, not an independently calibrated prediction score. Figure 4 reports GemNet raw and mean-shifted MAEs in Ha, without IQR normalization, and Kendall correlations; exact coverage, offset magnitudes, and additional metrics are reported in SI Note 5. Detailed reconstruction and evaluation procedures are in SI Note 5.

Data and code availability

The code, processed target tables, split identifiers, evaluation summaries, figure source data, manuscript sources, and reproducibility checks for this work are available in the corrected release `v0.2.0-corrected` of the project repository [29]. The original tmQM and QM9 datasets are available from their respective providers [16, 17]. Raw structures, LMDB databases, model checkpoints, and full prediction arrays are not redistributed in this release; the included manifests document their provenance.

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Author contributions

K.M.G. conceived and designed the study, performed the computational experiments, analysed the data, prepared the figures, and wrote the original manuscript. I.N. contributed to data preparation, computational workflows, and interpretation. O.V. and S.H. supervised the project. O.V. contributed to study design, interpretation, and manuscript revision. All authors reviewed and approved the manuscript.

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Supplementary Information for The Mechanics of Delta Learning: Target Design for Generalizable Scientific Machine Learning

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Scope and Reporting Conventions

This supplement asks how target construction changes the difficulty of learning molecular energies and their transfer between chemical domains. SchNet and GemNet are evaluated on the same 13 target definitions, with independent GNN seeds 101, 102, and 103 for every target. Each run is evaluated on the tmQM test split and on QM9 using its own best-validation checkpoint. The 78 main runs are supplemented by four DFTB runs: two targets, two architectures, and seed 101 only.

Unless stated otherwise, errors are in Hartree (Ha), and uncertainty denotes the sample standard deviation of independently evaluated seed-level scores. Predictions are not averaged into an ensemble. A smaller error measures better performance on that target; it does not by itself explain the physical origin of the improvement.

DFTB is shown in magenta and marked † in every comparison figure. It is a supporting control, not a member of the primary target panel. DFTB is excluded from all pooled correlations, regression fits, target-ranking statistics, leave-one-target-out analyses, and primary-panel summaries. Its own single-run metrics and explicitly matched-subset comparisons are reported separately. No across-seed error bar is assigned to a single DFTB run.

The target-panel table defines the identifiers used in plots. BoE-lin means the linear composition reference; BoE-MLP denotes the alternative nonlinear composition residual. An unsubscripted BoE reference is linear. In particular, the BoE-MLP target is fitted directly from $E_{\mathrm{ref}}^{\mathrm{DFT}}$, not from the linear-BoE residual.

Table S1: Complete target and evaluation panel. Counts are molecules per seed; both architectures use the same counts. DFTB† rows are display-only controls for pooled analyses.

ID	Target	Seeds	Train	Validation	Test	QM9
T1	$E_{\mathrm{ref}}^{\mathrm{DFT}}$	3	56,830	7,104	7,104	130,807
T2	$E_{\mathrm{ref,BoE\text{-}lin}}^{\mathrm{DFT}}$	3	56,830	7,104	7,104	130,807
T3	$E_{\mathrm{ref,BoE\text{-}MLP}}^{\mathrm{DFT}}$	3	56,830	7,104	7,104	130,807
T4	$E_{\mathrm{ref}}^{\mathrm{xTB}}$	3	56,830	7,104	7,104	130,807
T5	$E_{\mathrm{ref,BoE\text{-}lin}}^{\mathrm{xTB}}$	3	56,830	7,104	7,104	130,807
T6	$\Delta_{\mathrm{ref}}^{\mathrm{xTB}}$	3	56,830	7,104	7,104	130,807
T7	$\Delta_{\mathrm{ref,BoE}}^{\mathrm{xTB}}$	3	56,830	7,104	7,104	130,807
T8	$\Delta_{\mathrm{lin}}^{\mathrm{BoB}}$	3	56,821	7,103	7,104	130,807
T9	$\Delta_{\mathrm{MLP}}^{\mathrm{BoB}}$	3	56,821	7,103	7,104	130,807
T10	$\Delta_{\mathrm{lin}}^{\mathrm{PRE}}$	3	56,821	7,103	7,104	128,988
T11	$\Delta_{\mathrm{MLP}}^{\mathrm{PRE}}$	3	56,821	7,103	7,104	128,988
T12	$\Delta_{\mathrm{lin}}^{\mathrm{PRE+A}}$	3	56,821	7,103	7,104	128,988
T13	$\Delta_{\mathrm{MLP}}^{\mathrm{PRE+A}}$	3	56,821	7,103	7,104	128,988
T14†	$E_{\mathrm{ref,BoE\text{-}lin}}^{\mathrm{DFTB\dagger}}$	1	56,540	7,068	7,068	130,797
T15†	$\Delta_{\mathrm{ref,BoE}}^{\mathrm{DFTB\dagger}}$	1	56,540	7,068	7,068	130,797

The architecture comparison is matched within every target: both architectures use its identical target values and split memberships. The entire panel does not have an identical training cohort across targets. The six descriptor residuals use 56,821 training and 7,103 validation molecules, whereas the other main targets use 56,830 and 7,104. All 13 main targets share the same 7,104 test molecules. DFTB has a separate success-filtered cohort.

Supplementary Section 1: Target Preprocessing and Scale Controls

Supplementary Note 1: Direct Preprocessing Controls

Composition referencing removes a large, learnable contribution to the atom-referenced DFT energy. With a linear BoE reference, test MAE falls from 0.019917 ± 0.000178 to 0.015648 ± 0.000077 Ha for SchNet and from 0.013360 ± 0.000423 to 0.009448 ± 0.000469 Ha for GemNet. The nonlinear BoE baseline narrows the residual further, but downstream errors increase to 0.024672 ± 0.000112 and 0.022380 ± 0.000452 Ha, respectively. A smaller residual distribution therefore does not guarantee an easier regression problem.

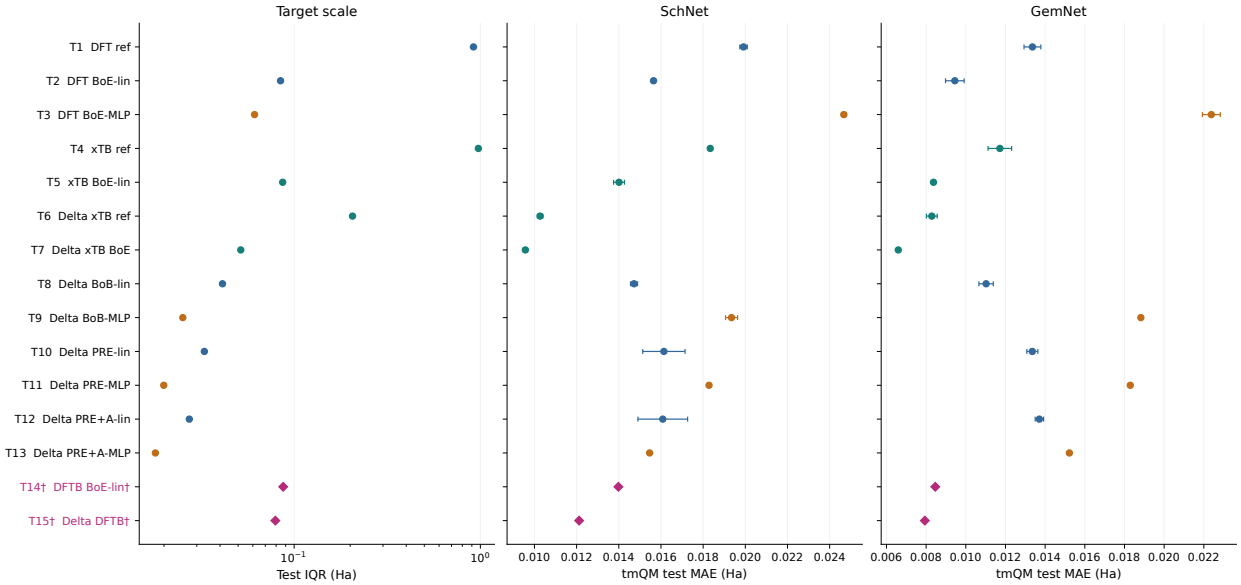


Figure S1: **Target scale and test error.** Test IQR and tmQM test MAE for all targets. Error bars are sample SD over three GNN seeds. Magenta DFTB† points are single-seed, reduced-cohort controls.

Table S2: Native-target test MAE (Ha), mean ± sample SD across seeds 101, 102, and 103. DFTB† has seed 101 only. xTB and DFTB direct-energy rows measure their own energies, not DFT reconstruction error.

Target	SchNet test MAE	GemNet test MAE
$E_{\mathrm{ref}}^{\mathrm{DFT}}$	0.019917 ± 0.000178	0.013360 ± 0.000423
$E_{\mathrm{ref,BoE\text{-}lin}}^{\mathrm{DFT}}$	0.015648 ± 0.000077	0.009448 ± 0.000469
$E_{\mathrm{ref,BoE\text{-}MLP}}^{\mathrm{DFT}}$	0.024672 ± 0.000112	0.022380 ± 0.000452
$E_{\mathrm{ref}}^{\mathrm{xTB}}$	0.018337 ± 0.000099	0.011718 ± 0.000595
$E_{\mathrm{ref,BoE\text{-}lin}}^{\mathrm{xTB}}$	0.014011 ± 0.000258	0.008372 ± 0.000031
$\Delta_{\mathrm{ref}}^{\mathrm{xTB}}$	0.010269 ± 0.000128	0.008286 ± 0.000273
$\Delta_{\mathrm{ref,BoE}}^{\mathrm{xTB}}$	0.009569 ± 0.000101	0.006596 ± 0.000048
$\Delta_{\mathrm{lin}}^{\mathrm{BoB}}$	0.014724 ± 0.000167	0.011024 ± 0.000363
$\Delta_{\mathrm{MLP}}^{\mathrm{BoB}}$	0.019347 ± 0.000287	0.018826 ± 0.000056
$\Delta_{\mathrm{lin}}^{\mathrm{PRE}}$	0.016139 ± 0.001004	0.013355 ± 0.000278
$\Delta_{\mathrm{MLP}}^{\mathrm{PRE}}$	0.018278 ± 0.000070	0.018293 ± 0.000090
$\Delta_{\mathrm{lin}}^{\mathrm{PRE+A}}$	0.016086 ± 0.001178	0.013710 ± 0.000212
$\Delta_{\mathrm{MLP}}^{\mathrm{PRE+A}}$	0.015462 ± 0.000035	0.015222 ± 0.000006
$E_{\mathrm{ref,BoE\text{-}lin}}^{\mathrm{DFTB\dagger}}$	0.013980	0.008462
$\Delta_{\mathrm{ref,BoE}}^{\mathrm{DFTB\dagger}}$	0.012119	0.007936

Table S3: Downstream training, validation, and test MAE (Ha). Training MAE averages all logged optimizer-step MAEs during epoch 80 as the weights evolve. Validation MAE is the best recorded validation score for each run, and test MAE is evaluated with that validation-selected checkpoint. Thus the training column describes the final epoch, rather than inference with frozen weights. Entries are means and sample SDs across seeds 101, 102, and 103. DFTB controls report seed 101 only.

Target	Training (epoch 80)	Validation (selected)	Test (selected)
GemNet			
$E_{\mathrm{ref}}^{\mathrm{DFT}}$	0.00996 ± 0.00124	0.01310 ± 0.00052	0.01336 ± 0.00042
$E_{\mathrm{ref,BoE\text{-}lin}}^{\mathrm{DFT}}$	0.00179 ± 0.00037	0.00930 ± 0.00046	0.00945 ± 0.00047
$E_{\mathrm{ref,BoE\text{-}MLP}}^{\mathrm{DFT}}$	0.00131 ± 0.00002	0.02190 ± 0.00044	0.02238 ± 0.00045
$E_{\mathrm{ref}}^{\mathrm{xTB}}$	0.00946 ± 0.00033	0.01183 ± 0.00075	0.01172 ± 0.00059
$E_{\mathrm{ref,BoE\text{-}lin}}^{\mathrm{xTB}}$	0.00154 ± 0.00001	0.00853 ± 0.00006	0.00837 ± 0.00003
$\Delta_{\mathrm{ref}}^{\mathrm{xTB}}$	0.00409 ± 0.00027	0.00810 ± 0.00020	0.00829 ± 0.00027
$\Delta_{\mathrm{ref,BoE}}^{\mathrm{xTB}}$	0.00100 ± 0.00008	0.00653 ± 0.00012	0.00660 ± 0.00005
$\Delta_{\mathrm{lin}}^{\mathrm{BoB}}$	0.00102 ± 0.00005	0.01123 ± 0.00025	0.01102 ± 0.00036
$\Delta_{\mathrm{MLP}}^{\mathrm{BoB}}$	0.00077 ± 0.00033	0.01853 ± 0.00015	0.01883 ± 0.00006
$\Delta_{\mathrm{lin}}^{\mathrm{PRE}}$	0.00075 ± 0.00010	0.01380 ± 0.00020	0.01335 ± 0.00028
$\Delta_{\mathrm{MLP}}^{\mathrm{PRE}}$	0.00355 ± 0.00029	0.01817 ± 0.00012	0.01829 ± 0.00009
$\Delta_{\mathrm{lin}}^{\mathrm{PRE+A}}$	0.00071 ± 0.00003	0.01373 ± 0.00006	0.01371 ± 0.00021
$\Delta_{\mathrm{MLP}}^{\mathrm{PRE+A}}$	0.00050 ± 0.00019	0.01563 ± 0.00006	0.01522 ± 0.00001
$E_{\mathrm{ref,BoE\text{-}lin}}^{\mathrm{DFTB\dagger}}$	0.00177	0.01130	0.00846
$\Delta_{\mathrm{ref,BoE}}^{\mathrm{DFTB\dagger}}$	0.00128	0.00920	0.00794
SchNet			
$E_{\mathrm{ref}}^{\mathrm{DFT}}$	0.02286 ± 0.00087	0.01940 ± 0.00010	0.01992 ± 0.00018
$E_{\mathrm{ref,BoE\text{-}lin}}^{\mathrm{DFT}}$	0.00650 ± 0.00036	0.01533 ± 0.00006	0.01565 ± 0.00008
$E_{\mathrm{ref,BoE\text{-}MLP}}^{\mathrm{DFT}}$	0.00544 ± 0.00025	0.02377 ± 0.00015	0.02467 ± 0.00011
$E_{\mathrm{ref}}^{\mathrm{xTB}}$	0.02133 ± 0.00007	0.01833 ± 0.00006	0.01834 ± 0.00010
$E_{\mathrm{ref,BoE\text{-}lin}}^{\mathrm{xTB}}$	0.00686 ± 0.00015	0.01387 ± 0.00012	0.01401 ± 0.00026
$\Delta_{\mathrm{ref}}^{\mathrm{xTB}}$	0.01087 ± 0.00011	0.01007 ± 0.00015	0.01027 ± 0.00013
$\Delta_{\mathrm{ref,BoE}}^{\mathrm{xTB}}$	0.00377 ± 0.00029	0.00947 ± 0.00006	0.00957 ± 0.00010
$\Delta_{\mathrm{lin}}^{\mathrm{BoB}}$	0.00355 ± 0.00025	0.01443 ± 0.00012	0.01472 ± 0.00017
$\Delta_{\mathrm{MLP}}^{\mathrm{BoB}}$	0.00307 ± 0.00036	0.01893 ± 0.00006	0.01935 ± 0.00029
$\Delta_{\mathrm{lin}}^{\mathrm{PRE}}$	0.00289 ± 0.00020	0.01597 ± 0.00015	0.01614 ± 0.00100
$\Delta_{\mathrm{MLP}}^{\mathrm{PRE}}$	0.00198 ± 0.00020	0.01817 ± 0.00006	0.01828 ± 0.00007
$\Delta_{\mathrm{lin}}^{\mathrm{PRE+A}}$	0.00295 ± 0.00051	0.01547 ± 0.00006	0.01609 ± 0.00118
$\Delta_{\mathrm{MLP}}^{\mathrm{PRE+A}}$	0.00140 ± 0.00008	0.01587 ± 0.00006	0.01546 ± 0.00003
$E_{\mathrm{ref,BoE\text{-}lin}}^{\mathrm{DFTB\dagger}}$	0.00682	0.01400	0.01398
$\Delta_{\mathrm{ref,BoE}}^{\mathrm{DFTB\dagger}}$	0.00495	0.01170	0.01212

Direct xTB and DFTB energy-prediction rows describe how well each GNN learns the semiempirical energy itself. They are not DFT reconstruction errors. For a residual $\Delta^b = y - b$, in contrast, the stacked predictor $\hat{y} = b + \widehat{\Delta}^b$ satisfies

$$\hat{y} - y = \widehat{\Delta}^b - \Delta^b. \tag{S1}$$

Residual MAE and RMSE are exactly those of the reconstructed parent-energy prediction on the same molecules. This equivalence does not extend to R^2, Pearson correlation, or rank correlation without reconstructing the parent energy.

Supplementary Note 2: What Scale Normalization Removes

For $y' = ay + c$, with $a \neq 0$, the IQR and a consistently rescaled prediction error change by $|a|$. For a fixed molecular graph,

$$\mathrm{IQR}(y') = |a|\mathrm{IQR}(y), \quad D_{\mathrm{raw}}(y') = a^2 D_{\mathrm{raw}}(y), \quad D_{\mathrm{IQR}}(y') = D_{\mathrm{IQR}}(y). \tag{S2}$$

IQR-normalized roughness therefore removes a pure change of units or amplitude. A fitted baseline is not generally a uniform rescaling: it can change molecular ordering, tail structure, and local differences.

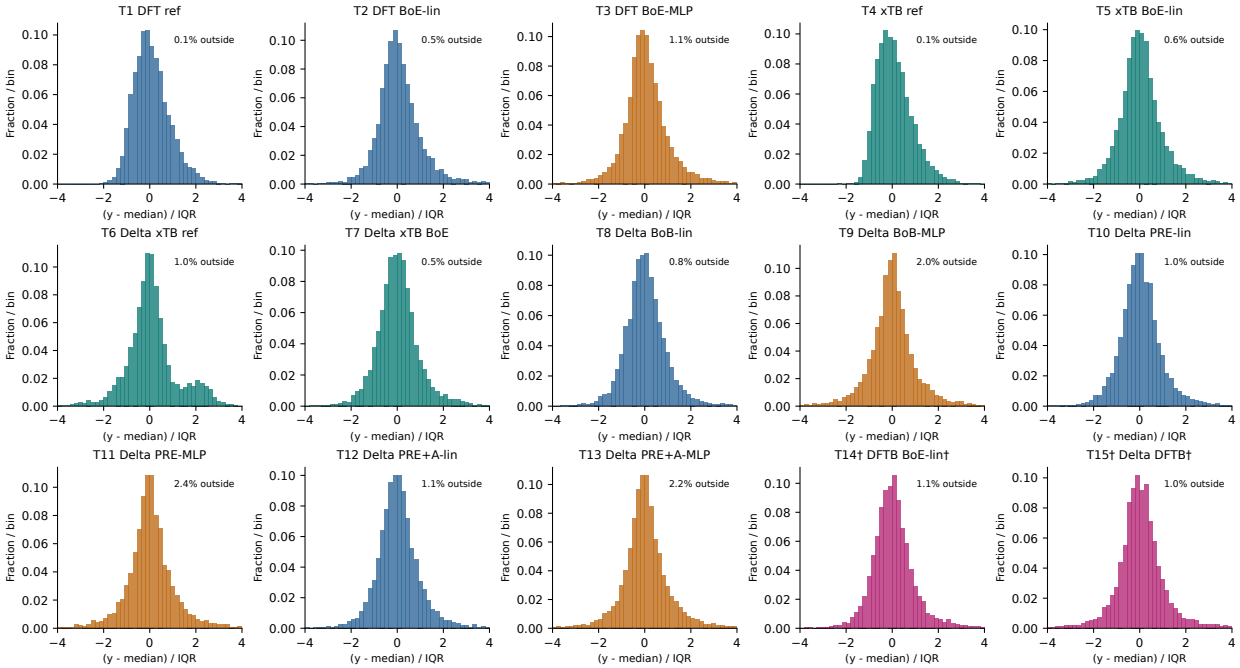


Figure S2: **Standardized test-target distributions.** Targets are centred by their test median and divided by their test IQR. Bin heights use the full test target population; annotations give fractions outside the displayed interval. DFTB† uses its valid subset.

This invariance is algebraic; it does not establish that independently optimized networks attain identical normalized errors. Nor does it eliminate correlations introduced by using a related scale factor on both diagnostic axes. Supplementary Note 7 examines that issue directly.

Supplementary Section 2: Baseline-Selection Diagnostics

Supplementary Note 3: Baseline Accuracy and Downstream Error

A useful baseline must do more than reproduce the parent energy accurately: the remaining signal must also be learnable by the downstream architecture. The baseline-score table reports frozen-baseline errors on test molecules. Test scores were calculated after fitting and did not select baseline parameters.

Table S4: Frozen baseline performance against the DFT-BoE parent (Ha). Errors and correlations refer to the baseline alone; test labels are used here only for post-training evaluation. DFTB† uses its valid subset.

Residual	Test MAE	Test RMSE	Pearson r	Spearman ρ
$\Delta^{\mathrm{BoB}}_{\mathrm{lin}}$	0.029315	0.047333	0.839	0.823
$\Delta^{\mathrm{BoB}}_{\mathrm{MLP}}$	0.021124	0.041346	0.881	0.901
$\Delta^{\mathrm{PRE}}_{\mathrm{lin}}$	0.024967	0.057663	0.771	0.893
$\Delta^{\mathrm{PRE}}_{\mathrm{MLP}}$	0.018746	0.068647	0.731	0.934
$\Delta^{\mathrm{PRE+A}}_{\mathrm{lin}}$	0.021128	0.047002	0.847	0.923
$\Delta^{\mathrm{PRE+A}}_{\mathrm{MLP}}$	0.016335	0.053602	0.811	0.950
$\Delta^{\mathrm{xTB}}_{\mathrm{ref,BoE}}$	0.035716	0.050873	0.830	0.787
$\Delta^{\mathrm{DFTB\dagger}}_{\mathrm{ref,BoE}}$	0.058773	0.086576	0.560	0.504

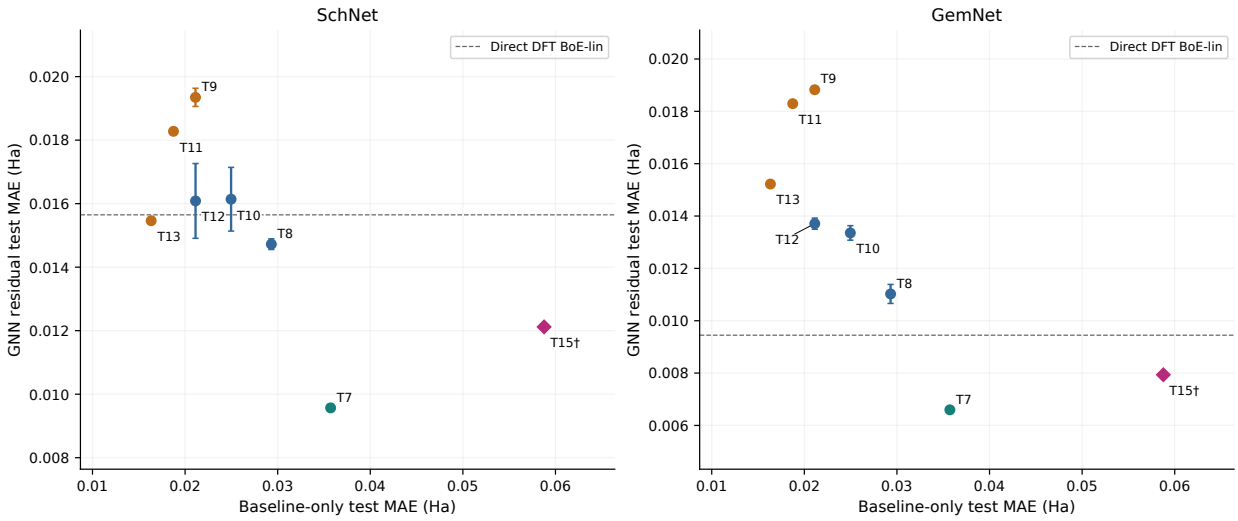


Figure S3: **Baseline accuracy versus downstream test error.** Each point pairs baseline-only test MAE with the GNN residual test MAE. Dashed lines show direct DFT-BoE learning. Error bars are GNN seed SD; DFTB† is a separate single-seed control.

More accurate descriptor baselines need not yield better downstream predictors. BoB, PRE, and PRE+A MLPs have smaller baseline errors than their corresponding linear models, yet none of their residual targets beats direct DFT-BoE learning for GemNet. For SchNet, linear BoB improves on direct learning, whereas linear PRE and PRE+A do not improve the mean test MAE. The latter two targets also have larger seed variability than most other SchNet targets.

The xTB BoE-referenced residual has the smallest mean tmQM test MAE in the 13-target panel for both architectures: 0.009569 ± 0.000101 Ha for SchNet and 0.006596 ± 0.000048 Ha for GemNet. Relative to direct DFT-BoE learning, these are 1.635-fold and 1.432-fold lower errors. This is an empirical target-design result, not proof of a particular electronic or geometric mechanism.

Supplementary Note 4: Descriptor Baseline Capacity

The linear and MLP descriptor models have the same parent target and, within each descriptor family, the same inputs and split memberships. Changing the fitting model tests baseline capacity without changing the DFT reference convention. Comparing descriptors under the common 512–128-hidden-unit MLP design tests information content, although input dimensions and total parameter counts differ.

MLP-PRE and MLP-PRE+A concentrate much of the residual near zero, but downstream errors and roughness do not decrease in proportion to the IQR. Their in-domain performance should not be extrapolated to QM9: fixed descriptor baselines can behave differently outside the tmQM domain. The transfer evaluation below tests this directly.

Supplementary Section 3: QM9 Transfer Controls

Supplementary Note 5: Complete Transfer Evaluation

Every main GNN seed is evaluated on QM9 without refitting the GNN or baseline. Native-target MAE is first reported on all molecules for which the target is defined: 130,807 for most targets, 128,988 for PRE and PRE+A residuals, and 130,797 for DFTB. Missing descriptor and DFTB rows are not imputed. Because cohorts differ, native-score tables document coverage but are not exact all-target, matched-cohort comparisons.

A second analysis reconstructs the common DFT-BoE parent and evaluates all main DFT-prediction routes on the same 128,988 molecules. Direct xTB and DFTB energy predictors are not DFT-prediction routes and do not enter these parent-energy tables. The displayed DFTB residual uses its 128,978-molecule intersection with this cohort. An additional exact-mask comparison with direct DFT and xTB appears in Supplementary Note 8.

For each seed, the oracle mean-shift diagnostic subtracts the mean prediction error on the evaluated QM9 cohort,

$$\hat{y}_i^{\text{shifted}} = \hat{y}_i - \frac{1}{N}\sum_{j=1}^{N}(\hat{y}_j - y_j). \tag{S3}$$

This uses QM9 reference labels and is not a deployable, uncalibrated prediction. It removes constant bias, but not ordering errors, heteroscedastic scatter, or outliers. It minimizes squared error over constant shifts, not necessarily MAE. Raw transfer errors remain essential.

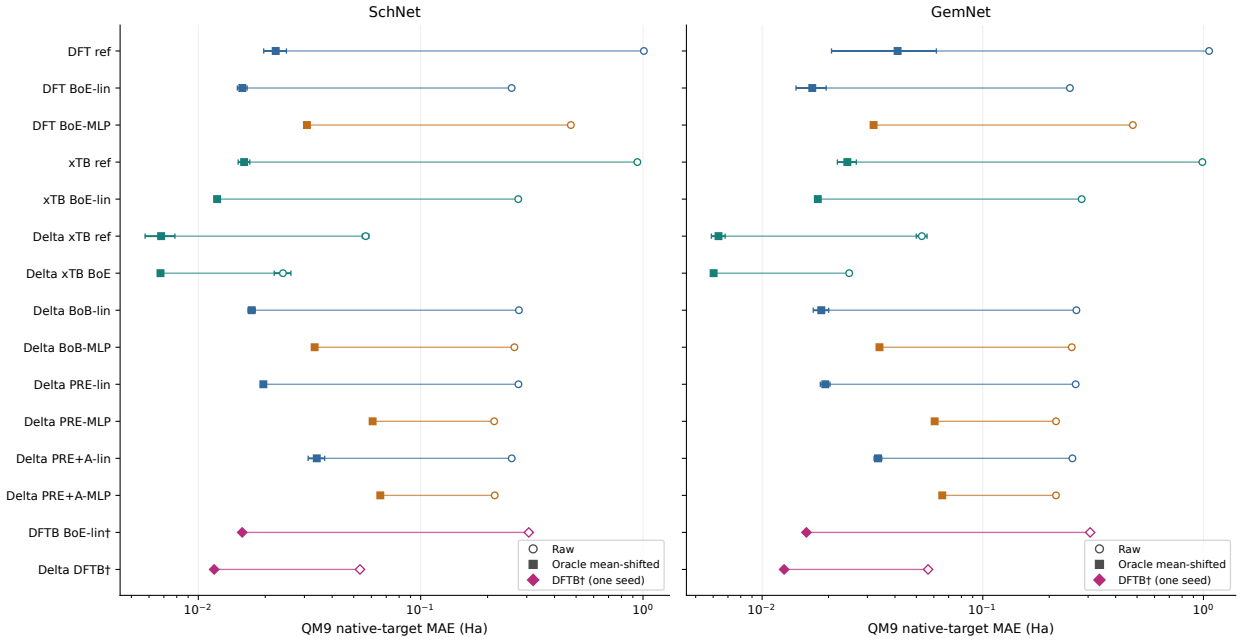


Figure S4: **Raw and oracle-shifted QM9 errors.** Open and filled symbols show native-target raw and shifted MAE; lines join the two diagnostics. Main results are three-seed means with sample SD. DFTB† is shown separately in magenta.

Table S5: SchNet QM9 native-target MAEs (Ha). Main targets: three independent seed scores, mean ± sample SD; DFTB†: seed 101. Native cohorts differ as listed. Shifts are estimated from the evaluated QM9 labels, not a separate calibration set.

Target	N	Raw MAE	Oracle-shift MAE
$E_{\mathrm{ref}}^{\mathrm{DFT}}$	130,807	1.009072 ± 0.005934	0.022291 ± 0.002609
$E_{\mathrm{ref,BoE\text{-}lin}}^{\mathrm{DFT}}$	130,807	0.256637 ± 0.004093	0.015778 ± 0.000802
$E_{\mathrm{ref,BoE\text{-}MLP}}^{\mathrm{DFT}}$	130,807	0.473662 ± 0.009143	0.030793 ± 0.000568
$E_{\mathrm{ref}}^{\mathrm{xTB}}$	130,807	0.942905 ± 0.001125	0.016076 ± 0.000972
$E_{\mathrm{ref,BoE\text{-}lin}}^{\mathrm{xTB}}$	130,807	0.274700 ± 0.002912	0.012149 ± 0.000225
$\Delta_{\mathrm{ref}}^{\mathrm{xTB}}$	130,807	0.056567 ± 0.001925	0.006805 ± 0.001043
$\Delta_{\mathrm{ref,BoE}}^{\mathrm{xTB}}$	130,807	0.024017 ± 0.002079	0.006760 ± 0.000192
$\Delta_{\mathrm{lin}}^{\mathrm{BoB}}$	130,807	0.276571 ± 0.002836	0.017383 ± 0.000645
$\Delta_{\mathrm{MLP}}^{\mathrm{BoB}}$	130,807	0.264029 ± 0.002472	0.033360 ± 0.000369
$\Delta_{\mathrm{lin}}^{\mathrm{PRE}}$	128,988	0.275298 ± 0.002464	0.019612 ± 0.000547
$\Delta_{\mathrm{MLP}}^{\mathrm{PRE}}$	128,988	0.214158 ± 0.000980	0.060797 ± 0.000118
$\Delta_{\mathrm{lin}}^{\mathrm{PRE+A}}$	128,988	0.256813 ± 0.001379	0.034096 ± 0.002906
$\Delta_{\mathrm{MLP}}^{\mathrm{PRE+A}}$	128,988	0.215363 ± 0.000066	0.065843 ± 0.000111
$E_{\mathrm{ref,BoE\text{-}lin}}^{\mathrm{DFTB\dagger}}$	130,797	0.306405	0.015751
$\Delta_{\mathrm{ref,BoE}}^{\mathrm{DFTB\dagger}}$	130,797	0.053349	0.011783

Table S6: GemNet QM9 native-target MAEs (Ha). Main targets: three independent seed scores, mean ± sample SD; DFTB†: seed 101. Native cohorts differ as listed. Shifts are estimated from the evaluated QM9 labels, not a separate calibration set.

Target	N	Raw MAE	Oracle-shift MAE
$E_{\mathrm{ref}}^{\mathrm{DFT}}$	130,807	1.060215 ± 0.018798	0.041115 ± 0.020489
$E_{\mathrm{ref,BoE\text{-}lin}}^{\mathrm{DFT}}$	130,807	0.248256 ± 0.006282	0.016878 ± 0.002639
$E_{\mathrm{ref,BoE\text{-}MLP}}^{\mathrm{DFT}}$	130,807	0.478472 ± 0.004472	0.031996 ± 0.000350
$E_{\mathrm{ref}}^{\mathrm{xTB}}$	130,807	0.988524 ± 0.003253	0.024332 ± 0.002402
$E_{\mathrm{ref,BoE\text{-}lin}}^{\mathrm{xTB}}$	130,807	0.280595 ± 0.003342	0.017877 ± 0.000265
$\Delta_{\mathrm{ref}}^{\mathrm{xTB}}$	130,807	0.052928 ± 0.002964	0.006344 ± 0.000454
$\Delta_{\mathrm{ref,BoE}}^{\mathrm{xTB}}$	130,807	0.024815 ± 0.000231	0.006029 ± 0.000117
$\Delta_{\mathrm{lin}}^{\mathrm{BoB}}$	130,807	0.265518 ± 0.002632	0.018540 ± 0.001494
$\Delta_{\mathrm{MLP}}^{\mathrm{BoB}}$	130,807	0.253024 ± 0.005363	0.034020 ± 0.000691
$\Delta_{\mathrm{lin}}^{\mathrm{PRE}}$	128,988	0.263566 ± 0.000964	0.019325 ± 0.000974
$\Delta_{\mathrm{MLP}}^{\mathrm{PRE}}$	128,988	0.214647 ± 0.000882	0.060508 ± 0.000347
$\Delta_{\mathrm{lin}}^{\mathrm{PRE+A}}$	128,988	0.254705 ± 0.002866	0.033508 ± 0.001345
$\Delta_{\mathrm{MLP}}^{\mathrm{PRE+A}}$	128,988	0.214623 ± 0.000429	0.065508 ± 0.000444
$E_{\mathrm{ref,BoE\text{-}lin}}^{\mathrm{DFTB\dagger}}$	130,797	0.306466	0.015873
$\Delta_{\mathrm{ref,BoE}}^{\mathrm{DFTB\dagger}}$	130,797	0.056495	0.012578

Parent-Energy Reconstruction and Trend Fidelity

Equation S1 allows raw MAE to be scored in residual space. Parent-energy R^2 and Kendall τ require reconstruction first. For E_{ref} and the nonlinear BoE route, the frozen linear DFT-BoE term is subtracted after reconstruction so that every route refers to the same parent quantity.

Table S7: SchNet transfer after reconstruction onto the common DFT-BoE parent. Primary routes use the identical 128,988-molecule descriptor-common QM9 cohort; DFTB† uses its 128,978-molecule intersection. RMSE and R^2 are means of seed-level oracle-shifted metrics. Kendall ordering is shift-invariant.

Route	Oracle MAE	RMSE	R^2	Kendall τ
$E_{\mathrm{ref}}^{\mathrm{DFT}}$	0.020707 ± 0.002107	0.037080	0.467	0.749 ± 0.008
$E_{\mathrm{ref,BoE\text{-}lin}}^{\mathrm{DFT}}$	0.015634 ± 0.000814	0.028811	0.681	0.746 ± 0.015
$E_{\mathrm{ref,BoE\text{-}MLP}}^{\mathrm{DFT}}$	0.030699 ± 0.000419	0.043151	0.285	0.549 ± 0.034
$\Delta_{\mathrm{ref}}^{\mathrm{xTB}}$	0.006603 ± 0.001017	0.009128	0.967	0.898 ± 0.013
$\Delta_{\mathrm{ref,BoE}}^{\mathrm{xTB}}$	0.006647 ± 0.000190	0.009381	0.966	0.895 ± 0.001
$\Delta_{\mathrm{lin}}^{\mathrm{BoB}}$	0.017229 ± 0.000583	0.030164	0.650	0.718 ± 0.012
$\Delta_{\mathrm{MLP}}^{\mathrm{BoB}}$	0.033290 ± 0.000453	0.045189	0.216	0.473 ± 0.009
$\Delta_{\mathrm{lin}}^{\mathrm{PRE}}$	0.019612 ± 0.000547	0.032431	0.596	0.687 ± 0.002
$\Delta_{\mathrm{MLP}}^{\mathrm{PRE}}$	0.060797 ± 0.000118	0.108187	-3.495	0.506 ± 0.006
$\Delta_{\mathrm{lin}}^{\mathrm{PRE+A}}$	0.034096 ± 0.002906	0.066634	-0.709	0.734 ± 0.002
$\Delta_{\mathrm{MLP}}^{\mathrm{PRE+A}}$	0.065843 ± 0.000111	0.117543	-4.306	0.710 ± 0.000
$\Delta_{\mathrm{ref,BoE}}^{\mathrm{DFTB\dagger}}$	0.011623	0.016650	0.893	0.810

Table S8: GemNet transfer after reconstruction onto the common DFT-BoE parent. Primary routes use the identical 128,988-molecule descriptor-common QM9 cohort; DFTB† uses its 128,978-molecule intersection. RMSE and R^2 are means of seed-level oracle-shifted metrics. Kendall ordering is shift-invariant.

Route	Oracle MAE	RMSE	R^2	Kendall τ
$E_{\mathrm{ref}}^{\mathrm{DFT}}$	0.039317 ± 0.018274	0.062145	-0.697	0.651 ± 0.029
$E_{\mathrm{ref,BoE\text{-}lin}}^{\mathrm{DFT}}$	0.016407 ± 0.002413	0.030401	0.643	0.726 ± 0.048
$E_{\mathrm{ref,BoE\text{-}MLP}}^{\mathrm{DFT}}$	0.031678 ± 0.000363	0.044438	0.242	0.547 ± 0.024
$\Delta_{\mathrm{ref}}^{\mathrm{xTB}}$	0.006311 ± 0.000461	0.008690	0.971	0.896 ± 0.009
$\Delta_{\mathrm{ref,BoE}}^{\mathrm{xTB}}$	0.005982 ± 0.000095	0.008393	0.973	0.901 ± 0.001
$\Delta_{\mathrm{lin}}^{\mathrm{BoB}}$	0.018193 ± 0.001493	0.032728	0.587	0.698 ± 0.030
$\Delta_{\mathrm{MLP}}^{\mathrm{BoB}}$	0.033881 ± 0.000583	0.045824	0.193	0.494 ± 0.029
$\Delta_{\mathrm{lin}}^{\mathrm{PRE}}$	0.019325 ± 0.000974	0.032730	0.588	0.703 ± 0.017
$\Delta_{\mathrm{MLP}}^{\mathrm{PRE}}$	0.060508 ± 0.000347	0.107942	-3.475	0.502 ± 0.008
$\Delta_{\mathrm{lin}}^{\mathrm{PRE+A}}$	0.033508 ± 0.001345	0.066759	-0.712	0.738 ± 0.007
$\Delta_{\mathrm{MLP}}^{\mathrm{PRE+A}}$	0.065508 ± 0.000444	0.117164	-4.272	0.711 ± 0.001
$\Delta_{\mathrm{ref,BoE}}^{\mathrm{DFTB\dagger}}$	0.012526	0.018161	0.873	0.790

On the matched main cohort, both xTB residual formulations retain substantially smaller oracle-shifted errors than the descriptor-MLP residuals. The BoE-referenced xTB route gives approximately 0.00665 Ha for SchNet and 0.00598 Ha for GemNet, with Kendall τ near 0.895 and

0.901. The atom-referenced xTB residual is also competitive; composition referencing does not uniformly improve every transfer metric for both architectures.

MLP-PRE and MLP-PRE+A have oracle-shifted errors near 0.061–0.066 Ha and negative parent-energy R^2, despite small in-domain residual widths. PRE+A can retain a moderate Kendall ordering while having poor RMSE and R^2: rank fidelity does not protect against large amplitude errors or outliers. These failures limit claims that a flexible fitted baseline necessarily improves transfer.

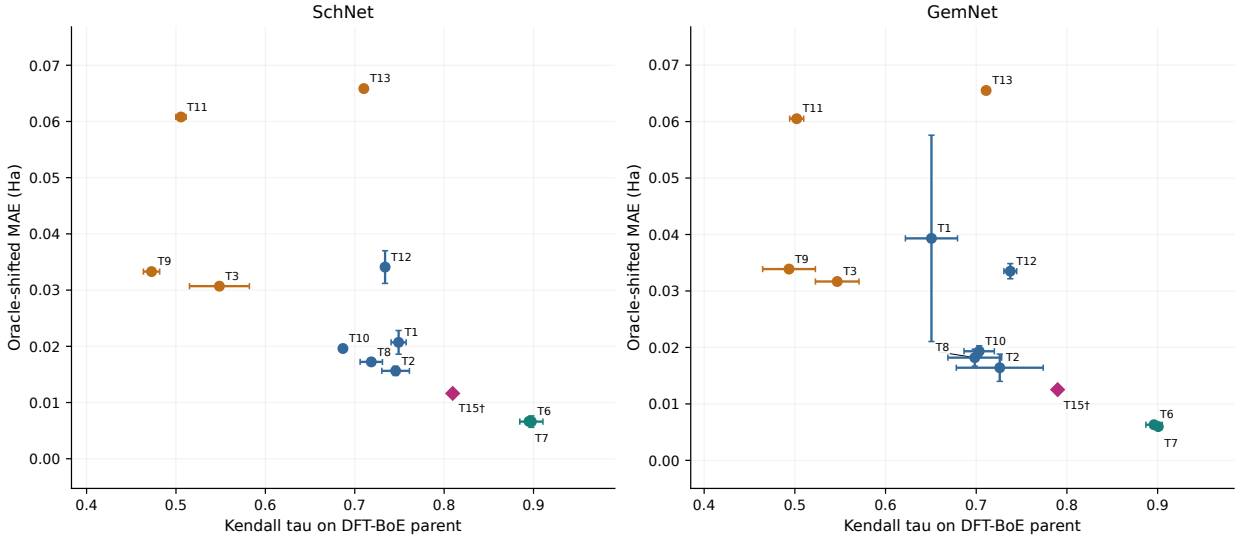


Figure S5: **Trend fidelity and remaining QM9 error.** Kendall τ versus oracle-shifted MAE after DFT-BoE reconstruction. Main routes use 128,988 common molecules; DFTB† uses its 128,978-molecule intersection and is not pooled with the main routes.

Supplementary Section 4: Architecture Controls and Roughness

Supplementary Note 6: Matched Architecture Panels

The roughness analysis uses all 13 main targets for both architectures. SchNet's proxy is PRE, and GemNet's proxy is PRE+A. These are fixed geometric proxies, not learned network representations. Pairwise message passing can encode more complex environments through multiple layers; PRE is not an exact representation of everything SchNet can learn.

The primary graphs use the 7,104 test molecules shared by all 13 targets. Standardized descriptors define cosine $k = 10$ neighbours. Unique undirected edges receive locally scaled Gaussian weights. For target y,

$$D_{\mathrm{raw}}(y) = \frac{\sum_{\{i,j\}\in\mathcal{E}} w_{ij}(y_i - y_j)^2}{\sum_{\{i,j\}\in\mathcal{E}} w_{ij}}, \qquad D_{\mathrm{IQR}}(y) = \frac{D_{\mathrm{raw}}(y)}{\mathrm{IQR}_{\mathrm{test}}(y)^2}. \tag{S4}$$

The normalized response is $n\text{-MAE}_{\mathrm{test}} = \mathrm{MAE}_{\mathrm{test}}/\mathrm{IQR}_{\mathrm{test}}(y)$. Test error, its scale denominator, and roughness use the same test cohort. Molecular descriptors alone define graph edges and weights; test labels supply the target differences and IQR after models are frozen. Validation is used for model selection and learning-rate scheduling, not the reported performance or roughness. This is a retrospective cross-target diagnostic, not a prospectively validated model-selection rule; using these test results to select future models would require a new independent evaluation set.

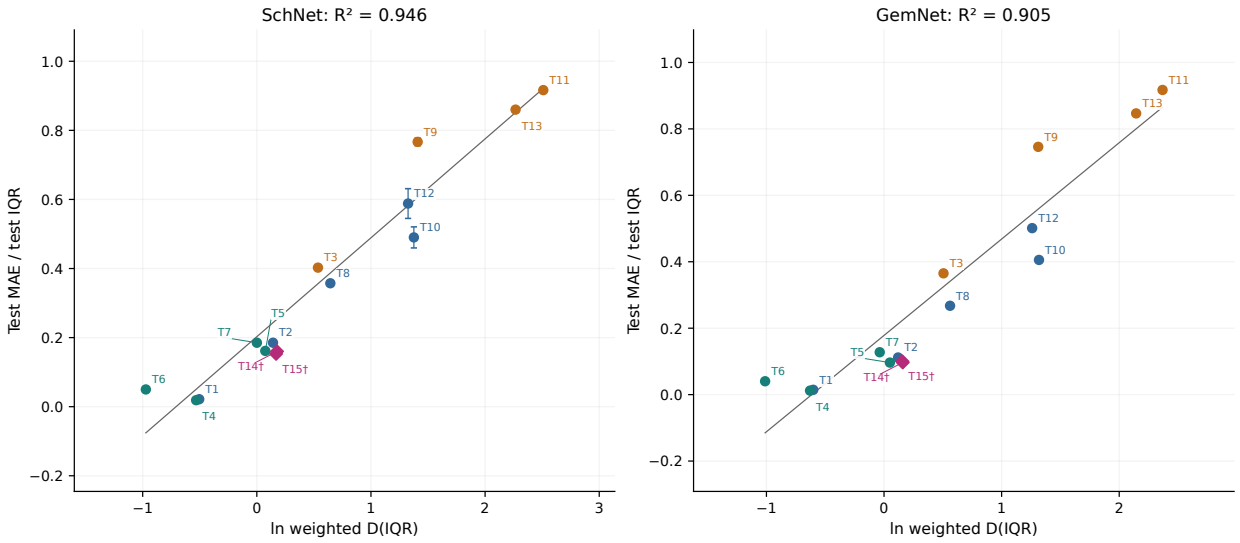


Figure S6: **Weighted test roughness versus normalized test error.** Lines fit the 13 main target means only. Error bars are three-seed SD divided by the fixed test IQR. DFTB† uses valid-subset test graphs and is excluded from fits and R^2.

Table S9: Architecture-proxy roughness and normalized test errors. Primary graph: 7,104 common test molecules; DFTB† graph: 7,068 DFTB-valid test molecules. Each error and roughness uses its matching test IQR. DFTB positions are visual controls, not members of the regression.

Target	Graph IQR (Ha)	PRE D_{IQR}	PRE+A D_{IQR}	SchNet n-MAE	GemNet n-MAE
$E_{\mathrm{ref}}^{\mathrm{DFT}}$	0.918321	0.6036	0.5477	0.0217	0.0145
$E_{\mathrm{ref,BoE\text{-}lin}}^{\mathrm{DFT}}$	0.084504	1.1522	1.1271	0.1852	0.1118
$E_{\mathrm{ref,BoE\text{-}MLP}}^{\mathrm{DFT}}$	0.061288	1.7086	1.6583	0.4026	0.3652
$E_{\mathrm{ref}}^{\mathrm{xTB}}$	0.975633	0.5875	0.5330	0.0188	0.0120
$E_{\mathrm{ref,BoE\text{-}lin}}^{\mathrm{xTB}}$	0.086765	1.0771	1.0512	0.1615	0.0965
$\Delta_{\mathrm{ref}}^{\mathrm{xTB}}$	0.205796	0.3782	0.3637	0.0499	0.0403
$\Delta_{\mathrm{ref,BoE}}^{\mathrm{xTB}}$	0.051679	1.0005	0.9640	0.1852	0.1276
$\Delta_{\mathrm{lin}}^{\mathrm{BoB}}$	0.041208	1.9043	1.7533	0.3573	0.2675
$\Delta_{\mathrm{MLP}}^{\mathrm{BoB}}$	0.025236	4.0925	3.7107	0.7666	0.7460
$\Delta_{\mathrm{lin}}^{\mathrm{PRE}}$	0.032934	3.9588	3.7353	0.4901	0.4055
$\Delta_{\mathrm{MLP}}^{\mathrm{PRE}}$	0.019944	12.3036	10.6818	0.9165	0.9172
$\Delta_{\mathrm{lin}}^{\mathrm{PRE+A}}$	0.027351	3.7630	3.5243	0.5881	0.5012
$\Delta_{\mathrm{MLP}}^{\mathrm{PRE+A}}$	0.017979	9.6504	8.5331	0.8600	0.8467
$E_{\mathrm{ref,BoE\text{-}lin}}^{\mathrm{DFTB\dagger}}$	0.087331	1.1936	1.1725	0.1601	0.0969
$\Delta_{\mathrm{ref,BoE}}^{\mathrm{DFTB\dagger}}$	0.079243	1.1839	1.1693	0.1529	0.1001

The descriptive semi-log fits are

$$n\text{-MAE}_{\mathrm{SchNet}} = 0.203 + 0.286 \ln D_{\mathrm{IQR}}, \qquad R^2 = 0.946, \tag{S5}$$

$$n\text{-MAE}_{\mathrm{GemNet}} = 0.178 + 0.290 \ln D_{\mathrm{IQR}}, \qquad R^2 = 0.905. \tag{S6}$$

Kendall τ is 0.872 for SchNet and 0.821 for GemNet over the 13 target means. These fits summarize the present panel; 39 GNN runs per architecture do not constitute 39 independent target-design observations.

Supplementary Note 7: Normalization and Panel Sensitivity

A high correlation between D_{IQR} and n-MAE can partly arise because both contain an inverse target IQR. We therefore repeat the analysis with unnormalized test MAE, unnormalized test roughness, and test-variance-normalized roughness. All variants use the same test graphs; only normalization changes. DFTB is excluded throughout.

Table S10: Cross-target associations with test error. S: SchNet; G: GemNet. Roughness is logarithm-transformed. DFTB is excluded from every row; N counts targets, not seeds. R^2 is in-sample, not cross-validated.

Comparison	N	S r	S R^2	S τ	G r	G R^2	G τ
Shared test IQR	13	0.972	0.946	0.872	0.951	0.905	0.821
Unnormalized error	13	0.245	0.060	0.231	0.576	0.332	0.487
Unnormalized roughness	13	-0.633	0.401	-0.564	-0.587	0.345	-0.590
Neither axis IQR-normalized	13	0.259	0.067	0.128	-0.137	0.019	-0.154
Variance-normalized roughness	13	0.436	0.190	0.410	0.334	0.111	0.282
DFT-BoE + seven residuals	8	0.966	0.933	0.929	0.954	0.909	0.786
Six descriptor residuals	6	0.919	0.844	0.867	0.905	0.820	0.733

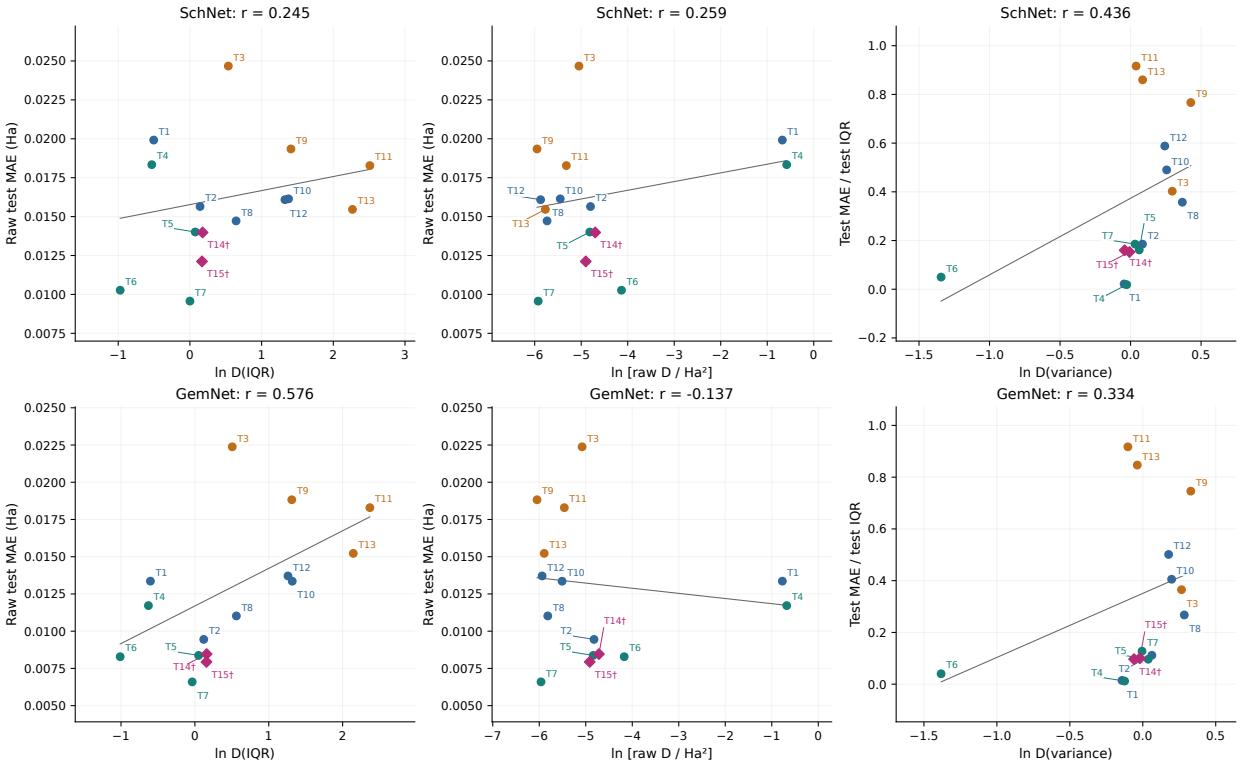


Figure S7: **Controls for shared normalization.** Columns use raw error with IQR-normalized roughness, neither axis IQR-normalized, and variance-normalized roughness with IQR-normalized error. Lines and Pearson r use only the 13 main targets. DFTB† remains a visual control.

The strongest association is not invariant to these choices. With raw error instead of n-MAE, R^2 falls to 0.060 for SchNet and 0.332 for GemNet. With neither axis IQR-normalized, R^2 is 0.067 and 0.019. The normalized fit must not be presented as equally strong evidence that unnormalized roughness predicts absolute error independently of target scale.

Using the same test cohort makes the comparison internally consistent, but does not eliminate shared-denominator coupling. As another descriptive check, we linearly residualize $\ln D_{\mathrm{IQR}}$ and $\ln n$-MAE against $\ln \mathrm{IQR}_{\mathrm{test}}$, including an intercept, and correlate the residual vectors. Partial Pearson correlations are 0.640 and 0.690. This checks association beyond a linear log-scale trend, but does not establish causality or a universal error law in a small, related target panel.

The association table also reports narrower panels: the DFT-BoE parent and its seven non-DFTB residuals, and the six descriptor residuals alone. Leaving out each main target in turn gives semi-log R^2 ranges of 0.927–0.965 for SchNet and 0.872–0.927 for GemNet. These are influence diagnostics, not cross-validated prediction scores.

Molecular neighbourhood information beyond shared normalization

A weaker raw correlation alone does not establish that the normalized association is artificial. We repeat the original control with 5,000 joint test-label permutations (seed 20260914), applying the same molecule permutation to all 13 targets and both architecture proxies. This preserves target distributions, IQRs, and cross-target relationships while disrupting label alignment with molecular neighbourhoods. Graphs, weights, and mean GNN test errors remain fixed; no GNN is retrained. DFTB is excluded. These controls are retrospective, not preregistered.

Table S11: **Corrected-test shared-normalization permutation control.** Fits use 13 primary targets per architecture. Null intervals are the 2.5th–97.5th percentiles over 5,000 joint molecule-label permutations, not confidence intervals for prediction. The one-sided p uses signed Pearson r, not R^2.

Model	Observed R^2	Null median R^2	Null 95% interval	p
GemNet	0.905	0.862	0.834–0.885	0.00020
SchNet	0.946	0.855	0.831–0.879	0.00020

For both models, the observed signed Pearson correlation of $\ln D_{\mathrm{IQR}}$ with test n-MAE exceeds all 5,000 shuffled values: one-sided plus-one $p = 1/5001 \approx 0.00020$, conditional on this panel and randomization scheme, without adjustment for the two architecture comparisons. High shuffled median R^2 values (GemNet: 0.862; SchNet: 0.855) show that target distributions and normalization contribute substantially. The stronger observed associations support additional information from label–geometry alignment beyond this conditional null, not a purely physical explanation or universal learnability law. The observed-minus-shuffled R^2 is not a decomposition of explained variance.

Table S12: **All similar-test-IQR primary target pairs.** The larger IQR is at most 1.10 times the smaller; selection does not use roughness or errors. Values follow target order. Roughness uses PRE for SchNet and PRE+A for GemNet; normalized error is mean test MAE/test IQR. These descriptive comparisons are not independent confirmation on new data.

Model	Targets	IQR (Ha)	D_{IQR}	n-MAE
GemNet	$E_{\mathrm{ref}}^{\mathrm{DFT}} \to E_{\mathrm{ref}}^{\mathrm{xTB}}$	0.91832 → 0.97563	0.548 → 0.533	0.015 → 0.012
GemNet	$E_{\mathrm{ref,BoE}}^{\mathrm{DFT}} \to E_{\mathrm{ref,BoE}}^{\mathrm{xTB}}$	0.08450 → 0.08677	1.127 → 1.051	0.112 → 0.096
GemNet	$\Delta_{\mathrm{MLP}}^{\mathrm{BoB}} \to \Delta_{\mathrm{lin}}^{\mathrm{PRE+A}}$	0.02524 → 0.02735	3.711 → 3.524	0.746 → 0.501
SchNet	$E_{\mathrm{ref}}^{\mathrm{DFT}} \to E_{\mathrm{ref}}^{\mathrm{xTB}}$	0.91832 → 0.97563	0.604 → 0.587	0.022 → 0.019
SchNet	$E_{\mathrm{ref,BoE}}^{\mathrm{DFT}} \to E_{\mathrm{ref,BoE}}^{\mathrm{xTB}}$	0.08450 → 0.08677	1.152 → 1.077	0.185 → 0.161
SchNet	$\Delta_{\mathrm{MLP}}^{\mathrm{BoB}} \to \Delta_{\mathrm{lin}}^{\mathrm{PRE+A}}$	0.02524 → 0.02735	4.092 → 3.763	0.767 → 0.588

The original 10% IQR-matching threshold selects three pairs, all concordant in roughness and normalized-error ordering for both models. Linear PRE+A has 8.4% larger IQR than MLP-BoB, but lower roughness and normalized error. Matching narrows, but does not eliminate, denominator differences. Together, these controls support a geometry-sensitive component while retaining the shared-normalization limitation; they do not establish causality.

Supplementary Section 5: DFTB Semiempirical Control

Supplementary Note 8: DFTB Coverage and Matched Evaluation

DFTB tests whether another semiempirical baseline can yield a useful residual. It remains a supporting control because successful calculations cover a different cohort and only seed 101 was trained. Of 71,038 neutral tmQM molecules, 362 lack successful DFTB calculations: 290 training, 36 validation, and 36 test molecules. Valid splits contain 56,540, 7,068, and 7,068 molecules. Ten QM9 DFTB failures leave 130,797 valid molecules. Failed energies are not imputed.

Raw DFTB calculations and fixed standard-molecule reference inputs were reused; the learned composition correction was newly fitted on successful tmQM training molecules only. Historical fitted residuals were not used. The DFT parent is the canonical train-only DFT-BoE target restricted to the valid mask. Frozen DFTB coefficients generate validation, test, and QM9 targets.

On common DFTB-valid test molecules, the residual's D_{IQR} is 1.184 in PRE and 1.169 in PRE+A, compared with 1.127 and 1.113 for the DFT parent under identical masks. This is a paired target-roughness comparison, not a comparison across different graph cohorts. The DFTB residual is slightly rougher than the parent in both proxies, despite improving matched-subset test MAE; lower normalized roughness is therefore not a necessary condition for lower absolute error. DFTB energy and residual GNNs reach test MAEs of 0.013980 and 0.012119 Ha for SchNet, and 0.008462 and 0.007936 Ha for GemNet. Direct DFTB energy and DFT residual remain different prediction tasks.

Table S13: Seed-101 DFT reconstruction on exactly matched DFTB-valid evaluation molecules (Ha). Shifted errors are oracle diagnostics. Training cohorts remain different: DFTB used only successful training molecules. These rows are not pooled into any primary analysis.

Dataset	Route	N	S raw	S shifted	G raw	G shifted
tmqm	$E^{\mathrm{DFT}}_{\mathrm{ref,BoE\text{-}lin}}$	7,068	0.015563	0.015623	0.009794	0.009832
tmqm	$\Delta^{\mathrm{xTB}}_{\mathrm{ref,BoE}}$	7,068	0.009499	0.009480	0.006538	0.006560
tmqm	$\Delta^{\mathrm{DFTB\dagger}}_{\mathrm{ref,BoE}}$	7,068	0.012119	0.012125	0.007936	0.007954
qm9	$E^{\mathrm{DFT}}_{\mathrm{ref,BoE\text{-}lin}}$	130,797	0.252018	0.016036	0.255477	0.019912
qm9	$\Delta^{\mathrm{xTB}}_{\mathrm{ref,BoE}}$	130,797	0.024872	0.006813	0.024622	0.006151
qm9	$\Delta^{\mathrm{DFTB\dagger}}_{\mathrm{ref,BoE}}$	130,797	0.053349	0.011783	0.056495	0.012578

The matched table evaluates seed-101 direct-DFT, xTB-residual, and DFTB-residual predictors on identical DFTB-valid test or QM9 molecules. This removes evaluation-mask differences, but does not equalize their training cohorts. No separately trained exact-DFTB-mask direct-DFT GNN is included in the completed 82-run matrix. The control therefore supports a qualified comparison, not an unrestricted DFTB ranking or a three-seed uncertainty estimate.

Supplementary Section 6: Independent-Seed Results

Supplementary Note 9: GNN Seed Variation

The following tables report every main-run test and raw QM9 MAE, not only a representative seed. Each seed has its own validation-selected checkpoint. For $S = 3$, a metric m_s is summarized by

$$\bar{m} = \frac{1}{S}\sum_s m_s, \qquad s_m = \sqrt{\frac{1}{S-1}\sum_s (m_s - \bar{m})^2}. \tag{S7}$$

This SD measures training-seed variation for a fixed dataset and target. It is not uncertainty from drawing new molecules, changing the baseline fitting seed, or selecting a different domain. Deterministic IQR and roughness values have no GNN-seed error bars.

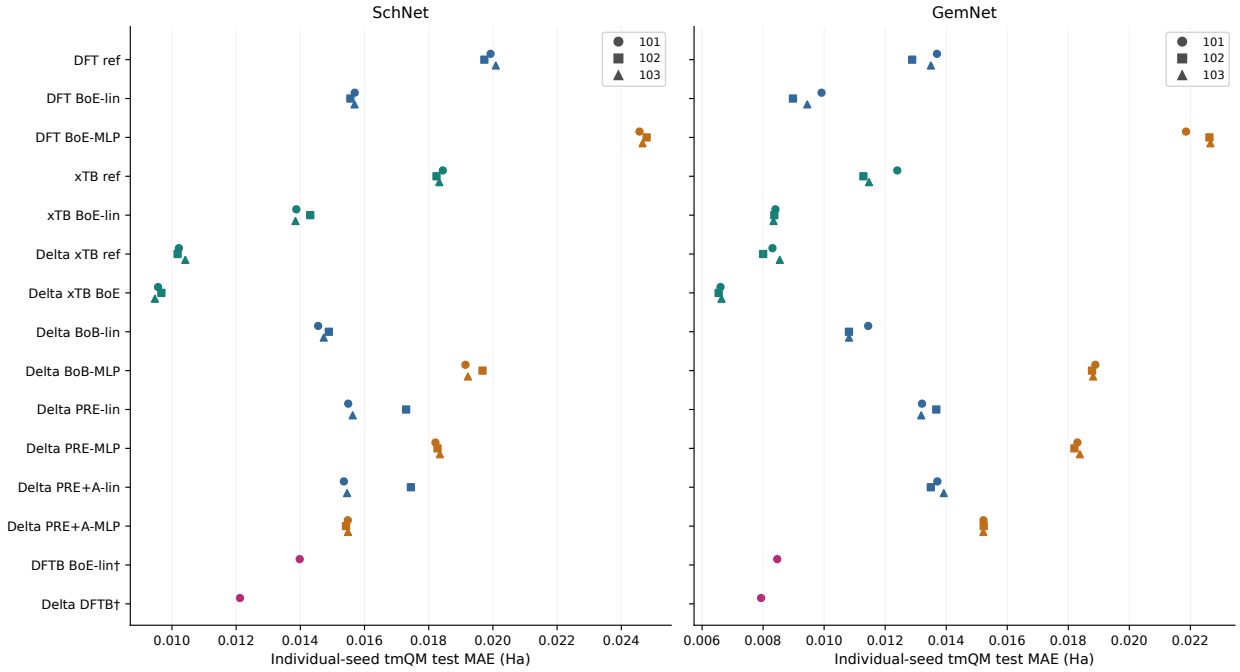


Figure S8: **Individual-seed tmQM test errors.** Circles, squares, and triangles show seeds 101, 102, and 103. All main targets have three runs per architecture. DFTB†, in magenta, has seed 101 only.

Table S14: SchNet individual-seed tmqm native-target MAEs (Ha). All predictions use each run's best-validation checkpoint. A dash denotes an intentionally unrun DFTB seed, not a zero error.

Target	101	102	103
$E_{\mathrm{ref}}^{\mathrm{DFT}}$	0.019928	0.019733	0.020089
$E_{\mathrm{ref,BoE\text{-}lin}}^{\mathrm{DFT}}$	0.015694	0.015559	0.015690
$E_{\mathrm{ref,BoE\text{-}MLP}}^{\mathrm{DFT}}$	0.024566	0.024789	0.024662
$E_{\mathrm{ref}}^{\mathrm{xTB}}$	0.018439	0.018242	0.018328
$E_{\mathrm{ref,BoE\text{-}lin}}^{\mathrm{xTB}}$	0.013876	0.014309	0.013849
$\Delta_{\mathrm{ref}}^{\mathrm{xTB}}$	0.010216	0.010176	0.010416
$\Delta_{\mathrm{ref,BoE}}^{\mathrm{xTB}}$	0.009568	0.009671	0.009469
$\Delta_{\mathrm{lin}}^{\mathrm{BoB}}$	0.014554	0.014888	0.014729
$\Delta_{\mathrm{MLP}}^{\mathrm{BoB}}$	0.019145	0.019675	0.019222
$\Delta_{\mathrm{lin}}^{\mathrm{PRE}}$	0.015492	0.017296	0.015631
$\Delta_{\mathrm{MLP}}^{\mathrm{PRE}}$	0.018210	0.018273	0.018350
$\Delta_{\mathrm{lin}}^{\mathrm{PRE+A}}$	0.015357	0.017444	0.015456
$\Delta_{\mathrm{MLP}}^{\mathrm{PRE+A}}$	0.015479	0.015422	0.015485
$E_{\mathrm{ref,BoE\text{-}lin}}^{\mathrm{DFTB\dagger}}$	0.013980	–	–
$\Delta_{\mathrm{ref,BoE}}^{\mathrm{DFTB\dagger}}$	0.012119	–	–

Table S15: GemNet individual-seed tmqm native-target MAEs (Ha). All predictions use each run's best-validation checkpoint. A dash denotes an intentionally unrun DFTB seed, not a zero error.

Target	101	102	103
$E_{\mathrm{ref}}^{\mathrm{DFT}}$	0.013696	0.012885	0.013498
$E_{\mathrm{ref,BoE\text{-}lin}}^{\mathrm{DFT}}$	0.009917	0.008979	0.009448
$E_{\mathrm{ref,BoE\text{-}MLP}}^{\mathrm{DFT}}$	0.021859	0.022624	0.022656
$E_{\mathrm{ref}}^{\mathrm{xTB}}$	0.012397	0.011287	0.011471
$E_{\mathrm{ref,BoE\text{-}lin}}^{\mathrm{xTB}}$	0.008404	0.008368	0.008343
$\Delta_{\mathrm{ref}}^{\mathrm{xTB}}$	0.008307	0.008003	0.008548
$\Delta_{\mathrm{ref,BoE}}^{\mathrm{xTB}}$	0.006606	0.006544	0.006638
$\Delta_{\mathrm{lin}}^{\mathrm{BoB}}$	0.011443	0.010813	0.010816
$\Delta_{\mathrm{MLP}}^{\mathrm{BoB}}$	0.018887	0.018777	0.018813
$\Delta_{\mathrm{lin}}^{\mathrm{PRE}}$	0.013208	0.013675	0.013182
$\Delta_{\mathrm{MLP}}^{\mathrm{PRE}}$	0.018298	0.018200	0.018381
$\Delta_{\mathrm{lin}}^{\mathrm{PRE+A}}$	0.013711	0.013497	0.013921
$\Delta_{\mathrm{MLP}}^{\mathrm{PRE+A}}$	0.015221	0.015229	0.015217
$E_{\mathrm{ref,BoE\text{-}lin}}^{\mathrm{DFTB\dagger}}$	0.008462	–	–
$\Delta_{\mathrm{ref,BoE}}^{\mathrm{DFTB\dagger}}$	0.007936	–	–

Table S16: SchNet individual-seed qm9 native-target MAEs (Ha). All predictions use each run's best-validation checkpoint. A dash denotes an intentionally unrun DFTB seed, not a zero error.

Target	101	102	103
$E_{\mathrm{ref}}^{\mathrm{DFT}}$	1.006303	1.005030	1.015885
$E_{\mathrm{ref,BoE\text{-}lin}}^{\mathrm{DFT}}$	0.252019	0.258079	0.259814
$E_{\mathrm{ref,BoE\text{-}MLP}}^{\mathrm{DFT}}$	0.483568	0.465546	0.471873
$E_{\mathrm{ref}}^{\mathrm{xTB}}$	0.942838	0.944062	0.941814
$E_{\mathrm{ref,BoE\text{-}lin}}^{\mathrm{xTB}}$	0.274410	0.277747	0.271944
$\Delta_{\mathrm{ref}}^{\mathrm{xTB}}$	0.055792	0.055150	0.058758
$\Delta_{\mathrm{ref,BoE}}^{\mathrm{xTB}}$	0.024874	0.025531	0.021646
$\Delta_{\mathrm{lin}}^{\mathrm{BoB}}$	0.273621	0.279276	0.276816
$\Delta_{\mathrm{MLP}}^{\mathrm{BoB}}$	0.261957	0.266765	0.263364
$\Delta_{\mathrm{lin}}^{\mathrm{PRE}}$	0.272982	0.277887	0.275026
$\Delta_{\mathrm{MLP}}^{\mathrm{PRE}}$	0.213049	0.214906	0.214518
$\Delta_{\mathrm{lin}}^{\mathrm{PRE+A}}$	0.258401	0.256111	0.255926
$\Delta_{\mathrm{MLP}}^{\mathrm{PRE+A}}$	0.215360	0.215299	0.215430
$E_{\mathrm{ref,BoE\text{-}lin}}^{\mathrm{DFTB\dagger}}$	0.306405	–	–
$\Delta_{\mathrm{ref,BoE}}^{\mathrm{DFTB\dagger}}$	0.053349	–	–

Table S17: GemNet individual-seed qm9 native-target MAEs (Ha). All predictions use each run's best-validation checkpoint. A dash denotes an intentionally unrun DFTB seed, not a zero error.

Target	101	102	103
$E_{\mathrm{ref}}^{\mathrm{DFT}}$	1.055482	1.044236	1.080928
$E_{\mathrm{ref,BoE\text{-}lin}}^{\mathrm{DFT}}$	0.255478	0.244060	0.245231
$E_{\mathrm{ref,BoE\text{-}MLP}}^{\mathrm{DFT}}$	0.476675	0.475178	0.483564
$E_{\mathrm{ref}}^{\mathrm{xTB}}$	0.986678	0.992279	0.986614
$E_{\mathrm{ref,BoE\text{-}lin}}^{\mathrm{xTB}}$	0.283163	0.281808	0.276816
$\Delta_{\mathrm{ref}}^{\mathrm{xTB}}$	0.055135	0.054089	0.049559
$\Delta_{\mathrm{ref,BoE}}^{\mathrm{xTB}}$	0.024624	0.025071	0.024750
$\Delta_{\mathrm{lin}}^{\mathrm{BoB}}$	0.262610	0.266210	0.267735
$\Delta_{\mathrm{MLP}}^{\mathrm{BoB}}$	0.257258	0.254821	0.246993
$\Delta_{\mathrm{lin}}^{\mathrm{PRE}}$	0.262614	0.263540	0.264543
$\Delta_{\mathrm{MLP}}^{\mathrm{PRE}}$	0.213722	0.214740	0.215478
$\Delta_{\mathrm{lin}}^{\mathrm{PRE+A}}$	0.253586	0.252567	0.257962
$\Delta_{\mathrm{MLP}}^{\mathrm{PRE+A}}$	0.214595	0.214208	0.215065
$E_{\mathrm{ref,BoE\text{-}lin}}^{\mathrm{DFTB\dagger}}$	0.306466	–	–
$\Delta_{\mathrm{ref,BoE}}^{\mathrm{DFTB\dagger}}$	0.056495	–	–

SchNet linear PRE and PRE+A show larger test variability than most other targets. Their seed-102 test MAEs are 0.017296 and 0.017444 Ha, compared with approximately 0.0154–0.0156 Ha for the other seeds. These runs are retained rather than discarded. Differences between mean scores should not be described as decisive when comparable to this variation.

Baseline Fitting Seeds Are Separate

A baseline-only check repeated each MLP fit with seeds 102 and 103, holding architecture, inputs, train-only preprocessing, and checkpoint selection fixed. Baseline seed 101 remains canonical; predictions and residual targets were not averaged over fitting seeds.

Table S18: Separate MLP-baseline fitting-seed check (Ha), seeds 101, 102, and 103. These are not the GNN seeds. Only baseline seed 101 defines the targets used by the 78 main GNN runs.

Descriptor	Test MAE	Range
BOB	0.021219 ± 0.000092	0.021124–0.021307
BOE	0.045442 ± 0.000512	0.044907–0.045926
PRE	0.018639 ± 0.000120	0.018509–0.018746
PRE+A	0.016523 ± 0.000201	0.016335–0.016735

Baseline test MAE SD is below 0.0006 Ha in each family, but individual residual labels are not identical. Paired test residuals show rare large differences relative to the canonical fit, with maxima of 1.093 Ha for PRE and 0.866 Ha for PRE+A. These scores use the already stored, frozen test predictions without refitting or selecting among baseline seeds. The present GNN error bars do not quantify this additional variability. Robustness to baseline fitting seed would require downstream evaluation on those alternative targets.

Supplementary Section 7: Supplementary Methods

Datasets, References, and Splits

The neutral tmQM benchmark has 71,038 molecules. Geometries derive from tmQM, and DFT energies are at ωB97M-V/def2-SVPD [1, 2, 3]. QM9 DFT energies use the same level, evaluated with Psi4 [4]. QM9 supplies no fitted baseline or GNN parameters.

Frozen split identifiers were retained during target regeneration. Joins use molecule IDs or explicitly mapped SIDs; CSV, LMDB, and prediction-array orders are not assumed to coincide. The corrected registry records exact split-file checksums. A filename containing *full* denotes concatenated split rows evaluated with one frozen model, not a model fitted on all molecules.

The fixed method-specific atom-reference convention is

$$E^m_{\mathrm{ref}}(i) = E^m_{\mathrm{raw}}(i) - \sum_Z n_Z(i)\epsilon^m_Z. \tag{S8}$$

Here ϵ^m_Z belongs to an external reference convention, not a regression fitted to held-out molecular energies. Canonical DFT E_{ref} values were retained as authoritative input; this is distinct from reusing a historical molecular-label-fitted composition reference.

Train-Only Baseline Fitting

Linear BoE predicts atom-referenced energy from 44 elemental counts, with an intercept. Ridge regularization is selected by five-fold shuffled training-only cross-validation, split seed 42, and mean fold-held-out MAE. Standardization is refitted within each fold. Final coefficients use training data only and are frozen for validation, test, and transfer. DFT, xTB, and DFTB have separate BoE fits; DFTB uses only successful training molecules.

The six descriptor baselines predict $Y = E^{\mathrm{DFT}}_{\mathrm{ref,BoE\text{-}lin}}$, giving $\Delta^b = Y - \hat{E}_b$. BoB is a 4,950-dimensional RDKit typed-bond-count descriptor, not conventional Coulomb-matrix Bag-of-Bonds. PRE has 34,650 pair-radial features, 35 Gaussian channels, and a 6.0 Å cutoff. Centres span 0.8–3.0 Å at 0.1 Å intervals (width 0.12 Å), then 3.25–6.0 Å at 0.25 Å intervals (width 0.30 Å). PRE+A adds 11,088 angular features, giving 45,738 total: 44 central elements, 21 unordered pairs of six radial shells, and 12 $\cos\theta$ bins.

Linear descriptor models remove training-zero-variance channels and divide by training population SD without centring. Filtering and scaling are refitted in every cross-validation fold. The descriptor ridge grid has 25 logarithmic values from 10^{-6} to 10^{6}; selected values must not be grid endpoints. Final fits use only the descriptor-common training split.

The four MLP baselines use hidden widths 512 and 128, SiLU, dropout 0.10, and a scalar linear output. MLP-BoE predicts $E^{\mathrm{DFT}}_{\mathrm{ref}}$ from counts; descriptor MLPs predict Y. AdamW uses learning rate 10^{-3}, weight decay 10^{-4}, SmoothL1 loss with $\beta = 0.10$, batch size 1024, clipping at 1, at most 800 epochs, and early-stopping patience 80 with minimum improvement 10^{-5} Ha. Training data fit parameters and preprocessing; validation selects the lowest-MAE checkpoint. Thus training-only fitting does not mean validation labels were never used for model selection.

Semiempirical Targets

GFN2-xTB supplies the xTB baseline [5]. Both referencing levels are included:

$$\Delta^{\mathrm{xTB}}_{\mathrm{ref}} = E^{\mathrm{DFT}}_{\mathrm{ref}} - E^{\mathrm{xTB}}_{\mathrm{ref}}, \tag{S9}$$

$$\Delta^{\mathrm{xTB}}_{\mathrm{ref,BoE}} = E^{\mathrm{DFT}}_{\mathrm{ref,BoE}} - E^{\mathrm{xTB}}_{\mathrm{ref,BoE}}. \tag{S10}$$

DFTB is analogous after success filtering. DFTB+ uses SCC-DFTB/DFTB2 with PTBP Slater–Koster parameters, without third-order or dispersion corrections [6, 7, 8]. This broad parameterization is a control, not a claim of optimal parameters for the molecular domain.

GNN Configurations and Selection

Both architectures use energy-only learning, a 6.0 Å cutoff, on-the-fly graphs, no periodic boundary conditions, and no force regression. SchNet uses 128 hidden channels, 128 filters, six interactions, 50 Gaussian functions, and sum readout [9]. GemNet-T uses three blocks, atom and edge widths 512, triplet and bilinear-triplet widths 64, radial and angular embedding widths 16, 128 radial functions, seven spherical functions, at most 50 neighbours, SiLU, and extensive summation [10]. It uses Gaussian radial functions, spherical harmonics, and a polynomial envelope with exponent five.

AdamW with AMSGrad starts at 5×10^{-4}. ReduceLROnPlateau uses factor 0.8 and patience three. Configurations specify MSE energy loss, EMA decay 0.999, clipping at 10, validation every 5,000 steps, and an 80-epoch budget. SchNet batch size is 16; GemNet batch size is four, including evaluation. Validation controls scheduling and checkpoint selection; test and QM9 labels do not.

Each target is standardized by its own training mean and population SD, shared between architectures and seeds. These were independently recalculated and checked against all 82 configurations, and are never fitted on validation, test, or QM9.

Table S19: Target scales in Ha. Training means and population standard deviations were independently checked against all 82 configuration files and remain unchanged. Test IQR normalizes reported test errors and test-graph roughness; it is not used for training.

Target	μ_{train}	σ_{train}	IQR_{test}
$E_{\text{ref}}^{\text{DFT}}$	1.035182	0.733244	0.918321
$E_{\text{ref,BoE-lin}}^{\text{DFT}}$	0.000000	0.091165	0.084504
$E_{\text{ref,BoE-MLP}}^{\text{DFT}}$	0.005657	0.065718	0.061288
$E_{\text{ref}}^{\text{xTB}}$	0.986222	0.764581	0.975633
$E_{\text{ref,BoE-lin}}^{\text{xTB}}$	0.000000	0.095927	0.086765
$\Delta_{\text{ref}}^{\text{xTB}}$	0.048960	0.251295	0.205796
$\Delta_{\text{ref,BoE}}^{\text{xTB}}$	0.000000	0.055191	0.051679
$\Delta_{\text{lin}}^{\text{BoB}}$	0.000000	0.051867	0.041208
$\Delta_{\text{MLP}}^{\text{BoB}}$	0.001850	0.029471	0.025236
$\Delta_{\text{lin}}^{\text{PRE}}$	-0.000000	0.040629	0.032934
$\Delta_{\text{MLP}}^{\text{PRE}}$	0.001371	0.011373	0.019944
$\Delta_{\text{lin}}^{\text{PRE+A}}$	-0.000000	0.033210	0.027351
$\Delta_{\text{MLP}}^{\text{PRE+A}}$	0.002017	0.010266	0.017979
$E_{\text{ref,BoE-lin}}^{\text{DFTB†}}$	0.000000	0.097167	0.087331
$\Delta_{\text{ref,BoE}}^{\text{DFTB†}}$	-0.000748	0.085898	0.079243

Roughness and Statistical Scope

Standardized roughness graphs remove zero-variance channels on the test graph cohort and divide retained channels by that cohort's population SD without centring. This unsupervised transformation is distinct from train-only regressor preprocessing. Cosine distances between normalized vectors define neighbours; unique undirected union edges receive

$$w_{ij} = \exp[-d_{ij}^2/(\sigma_i \sigma_j)], \tag{S11}$$

where σ_i is the tenth-neighbour distance. The common test graph has 51,535 PRE or 53,533 PRE+A edges. Every main target uses identical edges and weights within an architecture. DFTB and its paired DFT parent use separately constructed valid-subset test graphs, with 51,268 PRE or 53,246 PRE+A edges.

IQRs use linearly interpolated quartiles; variance normalization uses population variance. The primary fit is unweighted ordinary least squares across 13 target means, with an intercept and $\ln D_{\mathrm{IQR}}$ predictor. Pearson r, in-sample R^2, Spearman ρ, and Kendall τ are descriptive. Targets are related transformations of the same molecules, so target-wise independence is not assumed. An explicit 13-target whitelist excludes DFTB before every pooled calculation.

This corrected analysis reports two proxy graphs, normalization controls, panel-sensitivity checks, and 5,000 new joint test-label permutations with similar-IQR comparisons. Historical graph-grid results and synthetic-network rescaling fits are not presented as corrected-data experiments. The new permutation values use the corrected test protocol, not the historical validation results. The association does not establish a causal mechanism or guaranteed prospective ranking.

Prediction Audit and Reproducibility

The inventory contains 82 unique runs with complete predictions for both evaluation datasets. The SI audit re-read prediction arrays, verified unique IDs and complete available-target coverage, and recomputed all 164 native-target MAEs. The largest difference from the final ledgers was 6.7×10^{-16} Ha. All split target files matched corrected-registry checksums, split molecule sets were disjoint, and test and QM9 checkpoint paths matched. The final tmQM ledger records checkpoint checksums; the SI audit also records prediction and input checksums.

The SI bundle contains the target registry, final ledgers, target statistics, complete per-seed metrics, baseline scores, test-graph edges and weights, roughness values, correlation inclusion masks, checksums, and plotting code. Raw and oracle-shifted RMSE, R^2, Pearson correlation, Spearman correlation, Kendall correlation, and bias remain available in machine-readable tables where the printed tables select fewer metrics. These materials are distributed with the corrected code and data release, `v0.2.0-corrected` [11]. Raw structures, LMDB databases, model checkpoints, and full prediction arrays are not included in that release. Rebuilding this SI neither retrains models nor changes the main manuscript.

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